A revealed preference test for Choquet and Max-Min expected utility with ambiguity aversion

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Abstract

We develop a revealed preference test for the Choquet Expected Utility model with ambiguity aversion, which does not rely on specific functional form assumptions on the utility index. It is computationally efficient if the number of states is not too large, even for a large number of observations. This is a nice feature compared to other existing revealed preference tests for decision models with ambiguity. We illustrate the usefulness of our results by implementing our test on two experimental datasets from the literature, and we compare the empirical fit of this model to the subjective expected utility model.

JEL codes: C14, C60, D11, D12, D81

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In many situations, a decision-maker (DM) is unaware of the actual probabilities underlying the possible outcomes of her decision. In such cases, we say that the DM faces uncertainty. The standard theory for decision-making under uncertainty is Savage's subjective expected utility model (SEU) (Savage, 1954). In this model, the DM has a subjective probability measure over the various states of nature, and she acts as if she were an expected

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utility maximiser given these subjective probabilities. However, in his seminal contribution, Ellsberg (1961) demonstrated that many people exhibit ambiguity aversion in the sense that they prefer choice situations with known probabilities (risk) over choice situations where probabilities are not known (uncertainty). The SEU model cannot explain this phenomenon. Several models of decision-making under ambiguity have been developed as a response.[1] Two prominent models used to account for ambiguity aversion are the Choquet Expected Utility model (CEU) of Schmeidler (1989)[2] and the Max-Min Expected Utility model (MEU) of Gilboa and Schmeidler (1989). The CEU model represents a DM's information by a single non-additive 'probability measure' called a capacity. The crucial difference between a capacity and a standard probability measure is that capacities need not satisfy the additivity requirement.[3] In the MEU model, the DM's information set is represented by a set of plausible probability distributions over the states of nature, i.e. the set of priors. Being ambiguity averse, the DM picks from this set the probability distribution that provides the lowest expected utility when evaluating her decisions. As such, the DM chooses the action that maximizes her minimal expected utility, where the minimum is taken over all possible probability distributions in her set of priors.

As shown by Schmeidler (1989) and Gilboa and Schmeidler (1989), under ambiguity aversion, i.e. when the capacity is supermodular,[4] the CEU model coincides with a special case of the MEU model where the set of priors coincides with the core of the capacity.[5] Henceforth, we call this the Choquet-Max-min Expected Utility or CMEU model.

As it happens, the set of priors generated by the capacity under ambiguity aversion is a polymatroid. Our main contribution is to develop an efficient algorithm to find the probability distributions that effectively minimize expected utility given the decision maker's choice. Our algorithm uses insights from the literature on polymatroid theory, together with the supermodularity of the capacity. The algorithm is flexible and can be adapted to account for additional conditions on the Bernoulli utility index, such as concavity or smoothness.[6]

[1] See Machina and Siniscalchi (2014) for a variety of models that can explain ambiguity aversion

[2] The Choquet Expected Utility model is also commonly referred to as the Rank-Dependent Expected Utility model.

[3] In particular, for two disjoint events A and B it might not be the case that for a capacity μ, $\mu(A \cup B) = \mu(A) + \mu(B)$.

[4] Here, we stick to Schmeidler's original definition of ambiguity aversion. Other, more recent contributions also use different notions of ambiguity aversion.

[5] The core is the set of all probability distributions whose value is greater than or equal to the capacity for all events. See below for a formal definition.

[6] In this paper, we focus on the non-smooth setting. For the smooth case, we refer to an earlier version of this paper which is available from the authors upon request.

We develop an efficient revealed preference test to evaluate the empirical performance of the CMEU model. Revealed preference analysis was pioneered by Samuelson (1948) and Houthakker (1950) and further developed by Afriat (1967), Diewert (1973) and Varian (1982). Since then, it has been used to analyze a wide variety of decision-making models, including various models of choice under risk.[7] The revealed preference approach is attractive as it imposes no functional specification, like CARA or CRRA, on the underlying Bernoulli utility index. As such, the method is robust to such specification errors.

Ambiguity aversion models have attracted some attention in the recent revealed preference literature, but we are the first to provide a test for the CEU model with ambiguity aversion.

Bayer, Bose, Polisson, and Renou (2012) established a revealed preference characterization for a class of ambiguity-averse models, including the MEU model. They derive revealed preference tests for the MEU model that involve verifying a system of non-linear inequalities. Verifying such a non-linear system is a challenging problem, meaning the test is difficult to use in practice. In particular, the current practice of verifying the revealed preference conditions for the MEU model would entail conducting a grid search on the set of all priors. As the dimension of this grid increases with the number of states and observations, this procedure is limited to datasets with a low number of states and observations.[8] Our main result shows that under the CMEU specification, one can verify the revealed preference conditions using a grid search on the set of possible capacities. Although this grid is exponential in the number of states, our algorithm is only polynomial in the number of observations. As such, our test is efficient whenever the number of states is not too large. This feature is the case for most (experimental) datasets, which usually limit the number of ambiguous states to 2 or 3. In contrast, the number of observations can be as large as 50.[9]

Next, Chambers, Echenique, and Saito (2016a) derive a revealed preference test for the MEU model. Their result, however, limits the number of states to two.[10] Our test, on the other hand, imposes no such limits.

Finally, Polisson, Quah, and Renou (2020) developed a revealed preference procedure

[7]See, among others Varian (1983), Green and Srivastava (1986), Varian (1988), Green and Osband (1991), Choi, Fisman, Gale, and Kariv (2007), Diewert (2012) and Chambers, Liu, and Martinez (2016b).

[8]See also section 3.2 of Bayer, Bose, Polisson, and Renou (2012) for a discussion of the computational burden using such grid search.

[9]For example, the dataset of Ahn, Choi, Gale, and Kariv (2014), which we use in our empirical illustration, has 50 observations per subject while there are only two ambiguous states.

[10]They also consider settings with more than two states, but then they assume that the Bernoulli utility index is linear.

to analyze various models of choice under risk and uncertainty, including the MEU model. Their analysis focuses on relaxing the concavity restriction on the underlying Bernoulli utility index. Similar to Bayer, Bose, Polisson, and Renou (2012), they obtain a set of non-linear inequalities. In the main part of our paper, we stick to imposing concavity on the underlying Bernoulli function as it gives a set of revealed preference restrictions that are easier to analyze. However, the approaches are not necessarily incompatible. In Appendix B, we show how to use their characterization with the polymatroid approach to develop a revealed preference test for the CMEU model without imposing concavity on the utility index. The test amounts to verifying a set of linear inequalities (conditional on a grid search on the set of possible capacities).[11]

Most papers in the experimental literature on ambiguity aversion impose functional form assumptions to test their models, like CRRA or CARA. The main benefit of the revealed preference approach is that we abstain from imposing such restrictions. In our empirical application, we compare the revealed preference fit of the CMEU model with the SEU model. For our illustration, we use data from two choice experiments. The first dataset from Ahn, Choi, Gale, and Kariv (2014) is a portfolio choice experiment with one risky and two uncertain states. They show that most of their subjects' choices cannot be distinguished from SEU at a statistically significant level, although their tendency to hedge against ambiguity could also be explained by CMEU (among other models). Next, we use a dataset from Hey and Pace (2014), who asked subjects to bet on the color of a ball drawn from a bingo blower. Their paper analyses various models of risk and ambiguity to see which model fits the data best. However, they do not focus on the SEU, MEU or CEU models.

We implement our test on both data sets to compare the performance of the CMEU model vis à vis the SEU model. We compare both models based on the goodness-of-fit in terms of the Houtman-Maks index (Houtman and Maks, 1985) and their permissiveness using the Bronars power (Bronars, 1987). We do not see a big difference in fit between the CMEU and SEU models on both datasets. In particular, the CMEU seems to perform best when it is equivalent to the SEU model.

Section I presents the Choquet Max-Min Expected Utility model and gives a revealed preference characterization. Section II presents our algorithm and gives our main char-

[11] Unfortunately, in practice, the number of linear inequalities becomes quite large very quickly when the number of observations increases. This means that it becomes difficult to implement the test in settings with many observations. Therefore, we only illustrate it for the data in our second illustration. The test in the main part of the paper, which imposes concavity on the utility index, does not have this problem.

acterization. Section III gives a revealed preference test for the SEU model. Section IV provides our illustrations, and Section V contains the conclusion. All proofs and additional material are in the Appendix.

I The Choquet Max-min Expected Utility Model

We consider a setting with a finite set of n states $S = \{1, \ldots, n\}$. For a vector $x \in \mathbb{R}^n$ we write x_s for the s-th component of x and for a subset $S' \subseteq S$, we denote by $x_{S'}$ the sub-vector of x that ranges over the states $s \in S'$, i.e. $(x_s)_{s \in S'}$.[12]

We assume that there exists a unique set function $\mu : 2^S \to [0, 1]$, called a capacity, representing the DM's beliefs of the various events (subsets) of S. We assume that μ satisfies the following conditions:

1. Normalization: $\mu(S) = 1$ and $\mu(\emptyset) = 0$,
2. Monotonicity: if $A \subseteq B$, then $\mu(A) \leq \mu(B)$,
3. Supermodularity: for all events A and $B \subseteq S$, $\mu(A) + \mu(B) \leq \mu(A \cup B) + \mu(A \cap B)$.

Conditions (1) and (2) define μ to be a capacity. Condition (3) requires μ to be supermodular.[13] The condition relaxes the usual additivity condition for probability measures. In particular, it tells us that for some events, the agent might be unwilling or unable to allocate the entire mass $\mu(A \cup B) + \mu(A \cap B)$ to either events A or B because of ambiguity aversion.

Next, a bundle $q \in \mathbb{R}^n_+$ specifies for each state $s \in S$, a non-negative amount of money q_s obtained when state s occurs, i.e. q_s is the amount of an Arrow-Debreu security for state s. Finally, we use the notation $p \in \mathbb{R}^n_{++}$ to denote a vector of strictly positive prices for the securities.

According to the CEU model, a decision-maker endowed with a monotone Bernoulli utility index $u : \mathbb{R}_+ \to \mathbb{R}$ and faced with a price vector p, a total budget $m > 0$ and a capacity μ, will choose the bundle q that maximizes expected utility under a Choquet integral $\int_C$.

[12] We use the inequalities $x_{S'} \geq 0$ if $x_s \geq 0$ for all $s \in S'$, we use $x_{S'} > 0$ if $x_s \geq 0$ and $x_{S'} \neq 0$ and we use $x_{S'} \gg 0$ if $x_s > 0$ for all $s \in S$.

[13] In the literature, this condition is often called 'convexity'.

$$\max_{q\geq 0}\left\{\int_C u\big(q_s\big)d\mu\right\}$$
$$\text{subject to } \sum_{s\in S} p_s\cdot q_s \leq m,$$

To give an explicit expression of the Choquet integral, let q be a consumption vector and rearrange the states in S such that:

$$q_{(1)} \leq q_{(2)} \ldots \leq q_{(n)}.$$

Note that for a monotone Bernoulli utility index: $u(q_{(1)}) \leq u(q_{(2)}) \leq \ldots \leq u(q_{(n)})$. Finally, let $A_{(i)}$ be the event $\{(i),\ldots,(n)\}$ with $A_{(n+1)} = \emptyset$. Then the Choquet integral $\int_C u(q_s)d\mu$ is defined as:

$$\int_C u(q_s)d\mu = \sum_{i=1}^{n}\left[\mu\left(A_{(i)}\right) - \mu\left(A_{(i+1)}\right)\right]\cdot u(q_{(i)})$$

As shown by Schmeidler (1989) and Gilboa and Schmeidler (1989), the CEU model with ambiguity aversion (i.e. with a supermodular capacity) is a special case of the Maxmin expected utility model (MEU) where the DM's set of priors coincides with the core $\Delta(\mu)$ of the capacity μ:

$$\Delta(\mu) = \{\pi\in\Pi : \forall E\subseteq S, \pi(E)\geq\mu(E)\},$$

where we denote by Π the set of all probability measures on S. In particular, if the DM is ambiguity averse (i.e. μ is supermodular), then:

$$\int_C u(q_s)d\mu = \min_{\pi\in\Delta(\mu)}\sum_{s\in S}\pi_s\cdot u(q_s).$$

As such, our agent, endowed with a Bernoulli utility index u and a supermodular capacity μ facing a price vector $p\gg 0$, will choose the bundle q that solves:

$$\max_{q\geq 0}\left\{\min_{\pi\in\Delta(\mu)}\sum_{s\in S}\pi_s\cdot u\big(q_s\big)\right\} \text{ subject to } \sum_{s\in S} p_s\cdot q_s\leq m.$$

We take this model as our model of interest and call it the Choquet-Max-min Expected utility or CMEU model.

For the revealed preference analysis, we start from a dataset $\mathcal{D} = (p^t, q^t, m^t)_{t\leq T}$ that

consists of a finite number of price vectors $(p^t)_{t\leq T}$, budgets $(m^t)_{t\leq T}$ and corresponding choices $(q^t)_{t\leq T}$. Throughout, we will denote by T the number of observations. For a dataset $\mathcal{D}$ to be well defined, we assume that $\sum_{s\in S} p_s^t \cdot q_s^t = m^t$ for all $t \leq T$, which means that the expenditure on the bundle q^t equals the income m^t.[14]

We say that the data set $\mathcal{D}$ is CMEU rationalizable for a given capacity μ if its observed choices result from rational choices according to the CMEU model for some (well-behaved) Bernoulli utility index.

Definition 1 (CMEU rationalizability)**.** *Let μ be a supermodular capacity. A data set $\mathcal{D} = (p^t, q^t, m^t)_{t\leq T}$ is CMEU rationalizable for μ if there exists a monotone and concave Bernoulli utility index $u : \mathbb{R}_+ \to \mathbb{R}$ such that for all observations $t \leq T$,*

$$q^t \in \arg\max_{q\geq 0} \left\{ \min_{\pi\in\Delta(\mu)} \sum_{s\in S} \pi_s \cdot u\big(q_s\big) \right\} \ \ s.t. \ \sum_{s\in S} p_s^t \cdot q_s \leq m^t.$$

Let us discuss the restrictions we impose on the underlying Bernoulli utility index in more detail. First, monotonicity requires that utility $u(q_s)$ increases with the value of q_s. This assumption is standard. Indeed, without this assumption, any dataset is rationalizable by the utility index that gives constant utility for all amounts. Next, concavity is also a standard assumption on the utility index. It captures the idea that the decision-maker is risk-averse. Polisson, Quah, and Renou (2020) developed a revealed preference test for various models of choice under uncertainty, including the MEU model (but not the CEU model), that abstains from imposing concavity restrictions on the utility index. Here, we impose it as it gives a set of revealed preference conditions that are easier to verify, and more efficient to test. In Appendix B, we use the results from Polisson, Quah, and Renou (2020) to develop a test for CMEU rationalizability without imposing concavity on the utility index.

The following preliminary result characterizes the class of all datasets $\mathcal{D}$ that are CMEU rationalizable.

Theorem 1. *A data set $\mathcal{D} = (p^t, q^t, m^t)_{t\leq T}$ is CMEU rationalizable for a given supermodular capacity μ if and only if there is a probability distribution $\pi^t \in \Delta(\mu)$ together with n-dimensional vectors $u^t \geq 0$, $\nu^t \gg 0$ and numbers $\lambda^t > 0$ such that:*

[14] If desired, we can therefore omit m^t from the dataset and set it equal to $\sum_{s\in S} p_s^t q_s^t$.

- *for all* $\pi \in \Delta(\mu)$,

$$\sum_{s \in S} \pi_s^t \cdot u_s^t \leq \sum_{s \in S} \pi_s \cdot u_s^t. \tag{RP-1}$$

- *for all observations* $t, k \leq T$ *and states* $s, s' \in S$,

$$u_s^t - u_{s'}^k \leq \nu_{s'}^k \cdot (q_s^t - q_{s'}^k). \tag{RP-2}$$

- *for all observations* $t \leq T$ *and states* $s \in S$,

$$\begin{aligned} \pi_s^t \cdot \nu_s^t &= \lambda^t \cdot p_s^t \ \textit{if}\ q_s^t > 0, \\ \pi_s^t \cdot \nu_s^t &\leq \lambda^t \cdot p_s^t \ \textit{if}\ q_s^t = 0. \end{aligned} \tag{RP-3}$$

Intuitively, the unknown numbers u_s^t correspond to the utility levels at the observed consumption quantities, q_s^t. Next, the values of ν_s^t correspond to the marginal utility levels $u'(q_s^t)$.[15]

The values of λ^t correspond to the shadow prices for the budget constraints. Finally, the probabilities π_s^t correspond to the probability distribution in $\Delta(\mu)$ that solves the MEU version of the CMEU model.

Given these interpretations, the first constraint (RP-1) reflects the restriction that the probability distribution π^t must minimize the expected utility over all possible distributions $\pi \in \Delta(\mu)$. The conditions (RP-2) follow from the concavity of the function u:

$$u(q_s^t) - u(q_{s'}^k) \leq u'(q_{s'}^k) \cdot (q_s^t - q_{s'}^k).$$

Condition (RP-3), in turn, is derived from the first-order Kuhn-Tucker conditions:

$$\pi_s^t \cdot u'(q_s^t) = \lambda^t \cdot p_s^t, \text{ when } q_s^t > 0,$$

and

$$\pi_s^t \cdot u'(q_s^t) \leq \lambda^t \cdot p_s^t, \text{ when } q_s^t = 0.$$

[15] To be more precise, if the utility index is non-differentiable, one has to replace marginal utilities with suitable sub or sup-differentials (Rockafellar, 1970). For a concave function, ∂f is a sup-differential of f at the point x if for all y, $u(y) - u(x) \leq \partial f \cdot (y - x)$. When the utility index is differentiable at x, the sup-differential is unique and coincides with the derivative. However, sup-differentials also exist at points where the Bernoulli utility index is not differentiable.

Theorem 1 shows that in order to rationalize a data set with the CMEU model, it suffices to find for every observation t, a probability distribution $\pi^t \in \Delta(\mu)$ and verify a collection of inequalities for unknown values of π^t, u^t, ν^t and λ^t.

In our application, we perform a grid search on the capacity μ. As in most datasets, the number of states is rather small; this is quite efficient. However, even if so, verifying the revealed preference conditions is still far from trivial since they involve terms that are quadratic in the unknowns. In particular, condition (RP-1) contains the quadratic terms $\pi_s^t \cdot u_s^t$ while condition (RP-3) contains the quadratic terms $\pi_s^t \cdot \nu_s^t$. However, if we could determine the probability distribution π^t, these conditions would become linear, and (RP-1)-(RP-4) could be verified with standard linear programming methods.

In the following sections, we will show that (conditional on the capacity μ) it is possible to assign particular values to the probabilities π^t and therefore transform the set of non-linear conditions (RP-1)-(RP-3) into a set of linear inequalities.

II The algorithm

For ease of reading, let us reiterate the definition of the core:

$$\Delta(\mu) = \{\pi \in \Pi : \forall E \subset S, \pi(E) \geq \mu(E)\}$$

It is useful to define the dual $V : 2^S \to [0, 1]$ given by $V(E) = 1 - \mu(S \setminus E)$. Then the core can be written as:

$$\Delta(\mu) \equiv \nabla(V) = \{\pi \in \Pi : \forall E \subset S, \pi(E) \leq V(E)\}$$

Observe that the function $V(.)$ gives an upper bound to all elements in the set $\Delta(\mu)$. It is easy to see that V is a normalized ($V(\emptyset) = 0$ and $V(S) = 1$), monotone (if $A \subseteq B$ then $V(A) \leq V(B)$) and submodular set function. The latter means that for all $A, B \subseteq S$:

$$V(A \cup B) + V(A \cap B) \leq V(A) + V(B).$$

In particular, the set $\nabla(V)$ is a polymatroid (Edmonds, 2003). Polymatroid theory is a widely studied topic in mathematics and optimization theory. It is closely connected to the class of greedy algorithms, as we will see shortly. Our algorithm takes two steps.

A Step 1

Let us first focus on condition (RP-1) which requires that π^t minimizes the expected utility:

$$\pi^t \in \arg\min_{\pi \in \nabla(V)} \sum_{s \in S} \pi_s \cdot u_s^t. \tag{1}$$

Given the definition of $\nabla(V)$, we can rewrite this as:

OP-0

$$\begin{aligned} \pi^t \in \arg\min_{\pi \in \Pi} & \sum_{s \in S} \pi_s \cdot u_s^t, \\ \text{s.t.} & \sum_{s \in A} \pi_s \leq V(A), \forall A \subseteq S, \\ & \sum_{s \in S} \pi_s = V(S) = 1. \end{aligned}$$

This is a linear programming problem. In particular, as V is submodular, it is a linear programming problem where the constraint set is a polymatroid. In general, this linear problem cannot be solved explicitly as we do not know the values of u_s^t in the objective function. However, when the set of restrictions forms a polymatroid, only the ranking of the different coefficients u_s^t matters to determine the optimal solutions. In particular, **OP-0** can be solved using a greedy algorithm (Edmonds, 2003).

For the algorithm, one starts with the empty set $S_0 = \emptyset$ and looks for a state in $s_{(1)} \in S \setminus S_0 = S$ with the lowest coefficient $u_{s_{(1)}}^t$ in the objective function (ties are broken arbitrarily). Then we define $S_1 = S_0 \cup \{s_{(1)}\} = \{s_{(1)}\}$ and define:

$$\widetilde{\pi}_{s_{(1)}}^t = V(S_1) - V(S_0) \equiv V(\{s_{(1)}\}).$$

In the next step, we look for a state $s_{(2)} \in S \setminus S_1$ with the second lowest coefficient $u_{s_{(2)}}^t$ in the objective function (again ties are broken arbitrarily), then we update $S_2 = S_1 \cup \{s_{(2)}\} = \{s_{(1)}, s_{(2)}\}$ and define:

$$\widetilde{\pi}_{s_{(2)}}^t = V(S_2) - V(S_1) = V(\{s_{(1)}, s_{(2)}\}) - V(\{s_{(1)}\}).$$

In general, at step m, we look for a state $s_{(m)} \in S \setminus S_{m-1}$, which has the m-th lowest coefficient $u_{s_{(m)}}^t$. Then, we update $S_m = S_{m-1} \cup \{s_{(m)}\} = \{s_{(1)}, \ldots, s_{(m)}\}$ and define:

$$\widetilde{\pi}_{s_{(m)}}^t = V(S_m) - V(S_{m-1}) = V(\{s_{(1)}, \ldots, s_{(m)}\}) - V(\{s_{(1)}, \ldots, s_{(m-1)}\}).$$

At each step, this algorithm requires us to pick the lowest coefficient u_s^t among the states $s \in S \setminus S_m$. Although the values u_s^t are not known, we can still do this as the values u_s^t are monotonic transformations of the observed quantities q_s^t:

$$q_s^t \geq q_{s'}^t \leftrightarrow u_s^t \geq u_{s'}^t.$$

Algorithm 1 outlines the procedure.

Algorithm 1 The greedy algorithm

1: Sort states $\{s_{(1)}, \ldots, s_{(n)}\} = S$ in ascending order according to the quantities q_s^t, i.e.

$$q_{s_{(1)}}^t \leq q_{s_{(2)}}^t \leq \ldots \leq q_{s_{(n)}}^t.$$

2: Initialize $S_0 = \emptyset$ and $i = 0$.
3: **for** $i = 1$ to n **do**
4: set $S_i = S_{i-1} \cup \{s_{(i)}\}$
5: set $\widetilde{\pi}_{s_{(i)}}^t = V(S_i) - V(S_{i-1})$
6: **end for**

It is not necessarily the case that the output $\widetilde{\pi}^t$ of Algorithm 1 is unique. Uniqueness only occurs if no two states have the same quantities q_s^t, as in this case, the ranking in step 1 will be unique.

In order to identify all possible outputs $\widetilde{\pi}^t$ of Algorithm 1, we need some additional notation. Let us partition the set S according to the values of q_s^t, say $I_1^t, \ldots, I_K^t$. So two states s, s' with equal values $q_s^t = q_{s'}^t$ belong to the same set. We order the sets $I_1^t, \ldots, I_K^t$ such that for $j < i$, $s \in I_j^t$ and $s' \in I_i^t$:

$$q_s^t < q_{s'}^t.$$

So I_1^t contains the set of states s with the lowest values of q_s^t, I_2^t includes the set of states with the second lowest values of q_s^t, and so on.

Define $L_i^t = \bigcup_{j<i} I_j^t$ as the set of states with values q_s^t below $q_{s'}^t$ where $s' \in I_i^t$ (here $L_1^t = \emptyset$). Then for any subset $A \subseteq I_i^t$ define:

$$\widetilde{V}_i^t(A) = V(A \cup L_i^t) - V(L_i^t).$$

$\widetilde{V}_i^t$ is called the L_i^t-contraction of V and it is also a submodular function. If $\widetilde{\pi}^t$ is a solution

to Algorithm 1, then for all i:[16]

$$\sum_{s \in A} \widetilde{\pi}_s^t \leq \widetilde{V}_i^t(A), \qquad \forall A \subseteq I_i^t, \tag{C.1}$$

$$\sum_{s \in I_i^t} \widetilde{\pi}_s^t = \widetilde{V}_i^t(I_i^t). \tag{C.2}$$

On the other hand, if $\widetilde{\pi}^t$ is such that (C.1)-(C.2) are satisfied, then it also gives a solution to the minimization problem **OP-0**.[17] As such, these conditions characterize the set of all possible outputs for Algorithm 1. This motivates the following Proposition.

Proposition 1. *Let $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ be a dataset that satisfies* (RP-1)-(RP-3) *given μ and let V be the dual of μ.*

- *If $(\pi^t, u^t, \nu^t, \lambda^t)_{t \leq T}$ is a solution to the revealed preference conditions* (RP-1)*,* (RP-2) *and* (RP-3)*, then it must be that for all $t \leq T$ and all i, the vector π^t satisfies conditions* (C.1) *and* (C.2) *(when replacing $\widetilde{\pi}^t$ by π^t).*

- *If the values $(\pi^t, u^t, \nu^t, \lambda^t)_{t \leq T}$ provide a solution to the revealed preference conditions* (RP-2) *and* (RP-3) *and if for all t and all i, the vector π^t satisfies* (C.1) *and* (C.2) *(when replacing $\widetilde{\pi}^t$ by π^t), then* (RP-1) *is also satisfied.*

The first part of the proposition shows that any solution π^t to the revealed preference conditions also satisfies conditions (C.1) and (C.2). The second part shows that if these conditions are satisfied for some distribution π^t, then verifying condition (RP-1) is no longer necessary. In other words, we may replace condition (RP-1) by conditions (C.1)-(C.2) when verifying the revealed preference conditions. As the functions $\widetilde{V}_i^t$ can be computed directly from the dataset $\mathcal{D}$ and the capacity μ, this effectively transforms the non-linear conditions (RP-1) into the linear inequalities in (C.1) and (C.2).

If for all states $s, s' \in S$ and all observations t, $q_s^t \neq q_{s'}^t$, Proposition 1 identifies the probabilities π^t that should be assigned to the various states. In this case, the solution to the greedy algorithm will give a unique value π_s^t to each state $s \in S$:

$$\pi_s^t = \widetilde{V}_i^t(I_i^t) = V(I_i^t \cup L_i^t) - V(L_i^t), \text{ where } I_i^t = \{s\}.$$

On the other hand, if for some states $s, s' \in S$ and some observation t, the quantities are

[16] See the proof of Proposition 1 for details.

[17] Again see the proof of Proposition 1 for details.

equal, e.g. $q_s^t = q_{s'}^t$, then Proposition 1 might not assign a unique value to the probabilities π_s^t and $\pi_{s'}^t$. This type of situation is handled in step 2 of our procedure.

B Step 2

Given Proposition 1, the problem of determining the probabilities over the states in I_i^t boils down to finding a suitable vector $\pi_{I_i^t}^t$ that satisfies the revealed preference conditions (RP-2) and (RP-3) as well as (C.1) and (C.2).

The inequalities in (RP-2) and (RP-3) that involve variables relating to the states in I_i^t for observation t are given by:

$$\begin{aligned}
&u_{s'}^k - u_s^t \leq \nu_s^t(q_{s'}^k - q_s^t), && && \forall t, k \in T, s' \in S, s \in I_i^t,\\
&\pi_s^t \nu_s^t = \lambda^t p_s^t && \text{if } q_s^t > 0, && \forall t \in T, s \in I_i^t\\
&\pi_s^t \nu_s^t \leq \lambda^t p_s^t && \text{if } q_s^t = 0 && \forall t \in T, s \in I_i^t.
\end{aligned}$$

Notice that q_s^t and u_s^t are the same for all $s \in I_i^t$. Now, assume for the moment that we know the values of $(u^k, \lambda^k)_{k \leq T}$. We would like to see which values of $\pi_{I_i^t}^t$ satisfy the above restrictions.

- If $q_s^t > 0$, we can substitute out ν_s^t to obtain:

$$\frac{\pi_s^t}{p_s^t}(u_{s'}^k - u_s^t) \leq \lambda^t (q_{s'}^k - q_s^t).$$

  If $q_s^t = q_{s'}^k$, then this restriction is always satisfied as, in this case, $u_s^t = u_{s'}^k$. If $q_s^t > q_{s'}^k$ then $u_s^t > u_{s'}^k$ and we obtain the equivalent restrictions,

$$\frac{\pi_s^t}{p_s^t} \geq \lambda^t \frac{q_s^t - q_{s'}^k}{u_s^t - u_{s'}^k} \text{ for all } q_{s'}^k < q_s^t.$$

  If $q_s^t < q_{s'}^k$ then $u_s^t < u_{s'}^k$, and we obtain the restrictions,

$$\frac{\pi_s^t}{p_s^t} \leq \lambda^t \frac{q_{s'}^k - q_s^t}{u_{s'}^k - u_s^t} \text{ for all } q_{s'}^k > q_s^t.$$

- If $q_s^t = 0$ we have the restrictions,

$$\frac{1}{\nu_s^t}(u_{s'}^k - u_s^t) \leq q_{s'}^k - q_s^t \text{ and } \frac{\pi_s^t}{\lambda^t p_s^t} \leq \frac{1}{\nu_s^t}.$$

From $q_s^t = 0$, we know that $u_s^t \leq u_{s'}^k$,[18] If also $q_{s'}^k = 0$, then $u_s^t = u_{s'}^k$ and the inequality on the left is automatically satisfied. If $q_{s'}^k > 0$ then $u_s^t < u_{s'}^k$ so we obtain that,

$$\frac{1}{\nu_s^t} \leq \frac{q_{s'}^k - q_s^t}{u_{s'}^k - u_s^t} \text{ for all } q_{s'}^k > q_s^t \text{ and } \frac{\pi_s^t}{\lambda^t p_s^t} \leq \frac{1}{\nu_s^t}.$$

If we substitute out $1/\nu_s^t$, and combine these two inequalities, we see that there exists a solution if and only if:

$$\frac{\pi_s^t}{p_s^t} \leq \lambda^t \frac{q_{s'}^k - q_s^t}{u_{s'}^k - u_s^t} \text{ for all } q_{s'}^k > q_s^t.$$

Summarizing, we can define:

$$UB_{I_i^t} = \min_{\{k \in T, s' \in S: q_{s'}^k > q_s^t\}} \lambda^t \frac{q_{s'}^k - q_s^t}{u_{s'}^k - u_s^t},$$
$$LB_{I_i^t} = \max_{\{k \in T, s' \in S: q_{s'}^k < q_s^t\}} \lambda^t \frac{q_s^t - q_{s'}^k}{u_s^t - u_{s'}^k},$$

where we define $LB_{I_i^t} = 0$ if the feasible set for the maximization problem is empty (e.g. $q_s^t = 0$) and $UB_{I_i^t} = \infty$ if the feasible set for the minimization is empty. Observe that the values of $UB_{I_i^t}$ and $LB_{I_i^t}$ are identical for all $s \in I_i^t$ (this follows from the fact that for $s, s' \in I_i^t$, $q_s^t = q_{s'}^t$ and $u_s^t = u_{s'}^t$). In other words, the restrictions of (RP-2) and (RP-3) on $\pi_{I_i^t}^t$ boil down to the requirement that,

$$LB_{I_i^t} \leq \frac{\pi_s^t}{p_s^t} \leq UB_{I_i^t}, \qquad \forall s \in I_i^t$$

Equivalently:

$$LB_{I_i^t} \leq \min_{s \in I_i^t} \frac{\pi_s^t}{p_s^t} \text{ and } \max_{s \in I_i^t} \frac{\pi_s^t}{p_s^t} \leq UB_{I_i^t}.$$

Consider the following two problems,

[18] This follows from the fact that $u_s^t - u_{s'}^k \leq \nu_{s'}^k (q_s^t - q_{s'}^k) = \nu_{s'}^k (-q_{s'}^k) \leq 0$.

$$
\begin{array}{llll}
\textbf{OP-I} & \min\limits_{\pi_{I_i^t}\geq 0} ub_{I_i^t} & \text{s.t. } \dfrac{\pi_s}{p_s^t} \leq ub_{I_i^t}, & \forall s \in I_i^t \\
 & & \sum\limits_{s\in A} \pi_s \leq \widetilde{V}_i^t(A) & \forall A \subseteq I_i^t, \\
 & & \sum\limits_{s\in I_i^t} \pi_s = \widetilde{V}_i^t(I_i^t), & \\
\textbf{OP-II} & \max\limits_{\pi_{I_i^t}\geq 0} lb_{I_i^t} & \text{s.t. } \dfrac{\pi_s}{p_s^t} \geq lb_{I_i^t}, & \forall s \in I_i^t \\
 & & \sum\limits_{s\in A} \pi_s \leq \widetilde{V}_i^t(A) & \forall A \subseteq I_i^t. \\
 & & \sum\limits_{s\in I_i^t} \pi_s = \widetilde{V}_i^t(I_i^t), &
\end{array}
$$

Notice that any solutions to either **OP-I** or **OP-II** also satisfies the conditions (C.1) and (C.2).

For the moment, assume that there exists a vector $\widehat{\pi}^t$ that provides a common solution to both **OP-I** and **OP-II** for I_i^t, across all i. In this case, if we have a rationalization $(\pi^t, u^t, \nu^t, \lambda^t)_{t\leq T}$ that satisfies (RP-1), (RP-2) and (RP-3), then from the reasoning above, we have that for all $s \in I_i^t$,

$$
LB_{I_i^t} \leq \min_{s\in I_i^t} \frac{\pi_s^t}{p_s^t} \leq \min_{s\in I_i^t} \frac{\widehat{\pi}_s^t}{p_s^t} \leq \max_{s\in I_i^t} \frac{\widehat{\pi}_s^t}{p_s^t} \leq \max_{s\in I_i^t} \frac{\pi_s^t}{p_s^t} \leq UB_{I_i^t}
$$

This follows from the fact that, from Proposition 1, the vector π^t must satisfy (C.1) and (C.2), so it is feasible for **OP-I** and **OP-II**. In other words, if we replace π^t by $\widehat{\pi}^t$, we will also obtain a rationalization for the same data set $\mathcal{D}$ (for the same values of $(u^t, \nu^t, \lambda^t)_{t\leq T}$).

To summarize, we have that $(\pi^t, u^t, \nu^t, \lambda_t)_{t\leq T}$ is a rationalization if and only if $(\widehat{\pi}^t, u^t, \nu^t, \lambda^t)_{t\leq T}$ is a rationalization. As such, to test if $\mathcal{D}$ is rationalizable, it suffices to verify (RP-1), (RP-2) and (RP-3) for the vector $\widehat{\pi}^t$ that simultaneously solves **OP-I** and **OP-II**.

Algorithm 2 shows how to compute such a common solution $\widehat{\pi}^t$. The algorithm can be thought of as a greedy way to maximize the lower bound of the values $\dfrac{\pi_s}{p_s^t}$ for all $s \in I_i^t$. It first uniformly increases this value for all states in $\Omega^0 = I_i^t$ until one of the inequalities in (Lm.4) is binding. Then it removes from Ω^0 all states for which this inequality is binding and tries to increment the values (uniformly) over the remaining states, and so on. The validity of step 4 in the algorithm, i.e. the fact that there exists a largest set (with respect to

set inclusion) for which condition (Lm.4) is binding, is a consequence of the submodularity of the function $\widetilde{V}_i^t$.

Algorithm 2 Max-Min

1: Set $m = 0$, and $\Omega^0 = I_i^t$.
2: **while** $\Omega^m \neq \emptyset$ **do**
3:

$$\varepsilon_{I_i^t}^m \in \arg\max_{\varepsilon_{I_i^t}^m} \sum_{s\in\Omega^m} \varepsilon_s^m \text{ subject to } \frac{\varepsilon_s^m}{p_s^t} = \frac{\varepsilon_{s'}^m}{p_{s'}^t} \qquad \forall s, s' \in \Omega^m \qquad \text{(Lm.1)}$$

$$\varepsilon_s^m = 0 \qquad \forall s \notin \Omega^m \qquad \text{(Lm.2)}$$

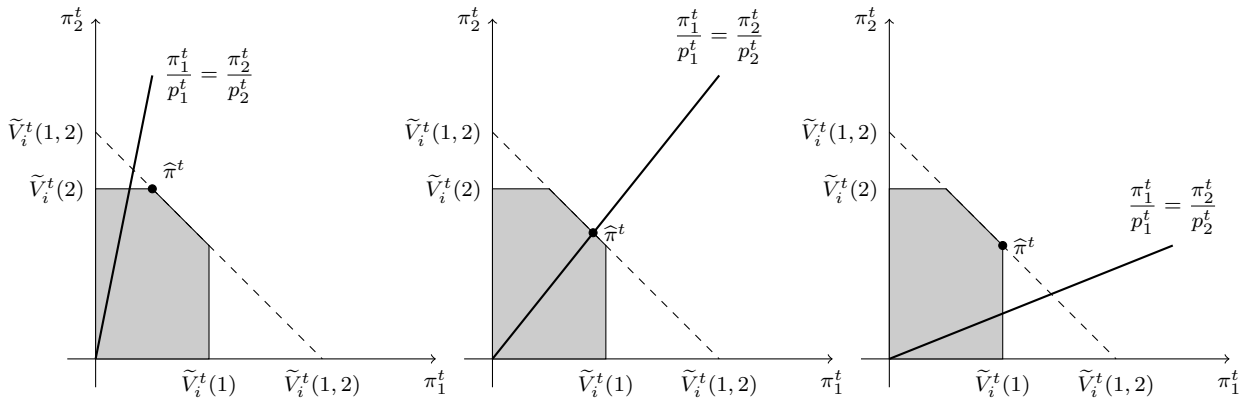
$$\varepsilon_s^m \geq 0 \qquad \forall s \in I_i^t \qquad \text{(Lm.3)}$$

$$\sum_{s\in A}\sum_{j=0}^{m} \varepsilon_s^j \leq \widetilde{V}_i^t(A) \qquad \forall A \subseteq I_i^t. \qquad \text{(Lm.4)}$$

4: Let A^m be the largest subset of I_i^t (in terms of the subset partial ordering) such that (Lm.4) is binding for the maximization problem in 3.
5: Set $\Omega^{m+1} = I_i^t \setminus A^m$
6: $m \leftarrow m + 1$.
7: **end while**
8: For all $s \in I_i^t$, set $\widehat{\pi}_s^t = \sum_{j=0}^m \varepsilon_s^j$.

It might be instructive to consider a simple example where I_i^t contains only two states, say 1 and 2. Figure 1 illustrates. The shaded area gives the values of π_1^t and π_2^t that satisfy the three constraints $\pi_1^t \leq \widetilde{V}_i^t(1)$, $\pi_2^t \leq \widetilde{V}_i^t(2)$ and $\pi_1^t + \pi_2^t \leq \widetilde{V}_i^t(1,2)$. The solid line through the origin gives the values of π_1^t and π_2^t that satisfy the condition $\frac{\pi_2^t}{p_2^t} = \frac{\pi_1^t}{p_1^t}$. Depending on the slope of this line (i.e. the value of $\frac{p_2^t}{p_1^t}$) three cases can be distinguished. If the slope is high enough, the algorithm will set $\widehat{\pi}_1^t$ and $\widehat{\pi}_2^t$ such that $\widehat{\pi}_2^t = \widetilde{V}_i^t(2)$ and $\widehat{\pi}_1^t + \widehat{\pi}_2^t = \widetilde{V}_i^t(1,2)$. The inequality $\widehat{\pi}_1^t \leq \widetilde{V}_i^t(1)$ will then be non-binding. In the middle case, when $\frac{p_2^t}{p_1^t}$ has moderate values, the algorithm will set $\widehat{\pi}_1^{\,t} + \widehat{\pi}_2^t = \widetilde{V}_i^t(1,2)$ and $\frac{\widehat{\pi}_2^t}{p_2^t} = \frac{\widehat{\pi}_1^t}{p_1^t}$, while the constraints $\widehat{\pi}_1^t \leq \widetilde{V}_i^t(1)$ and $\widehat{\pi}_2^t \leq \widetilde{V}_i^t(2)$ will both be non-binding. Finally, the third case, where $\frac{p_2^t}{p_1^t}$ is sufficiently low, will lead to a value where $\widehat{\pi}_1^t = \widetilde{V}_i^t(1)$ and $\widehat{\pi}_1^t + \widehat{\pi}_2^t = \widetilde{V}_i^t(1,2)$, while the constraint $\widehat{\pi}_2^t \leq \widetilde{V}_i^t(2)$ will be non-binding.

Figure 1: Illustrative example of Algorithm 2



The following proposition shows that Algorithm 2 is well-behaved and provides a common solution of both **OP-I** and **OP-II**.

Proposition 2. *Algorithm 2 stops in no more than $|I_i^t|$ iterations, and its solution $\widehat{\pi}^t$ solves both **OP-I** and **OP-II** simultaneously.*

The following Corollary summarizes the findings in Proposition 1 and 2.

Corollary 1. *Consider a dataset $\mathcal{D} = (q^t, p^t, m^t)_{t \leq T}$ and a supermodular capacity μ. Let $\widehat{\pi}^t$ be the solution of Algorithm 2.*

Then, the dataset $\mathcal{D}$ is CMEU rationalizable for μ if and only if for all $t \leq T$, there exists a vector $u^t \geq 0$, a vector $\nu^t \gg 0$ and a number $\lambda^t > 0$ such that $(\widehat{\pi}^t, u^t, \nu^t, \lambda^t)_{t \leq T}$ satisfies conditions (RP-2) *and* (RP-3)*.*

The following Theorem gives our main result. It uses the result from the previous Corollary to reformulate the conditions in Theorem 1 to a set of linear inequalities. Interestingly, these also no longer involve the unknowns u^t.

Theorem 2. *Let $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ be a dataset and let μ be a supermodular capacity. Let $\widehat{\pi}^t$ be the output from Algorithm 2.*

Then $\mathcal{D}$ is CMEU rationalizable for μ if and only if for all $t \leq T$, there exists a vector $\nu^t \gg 0$ and a number $\lambda^t > 0$ such that:

- *for all observations $t \in T$ and states $s \in S$*

$$\begin{aligned} \widehat{\pi}_s^t \cdot \nu_s^t &= \lambda^t \cdot p_s^t && \text{if } q_s^t > 0, \\ \widehat{\pi}_s^t \cdot \nu_s^t &\leq \lambda^t \cdot p_s^t && \text{if } q_s^t = 0. \end{aligned} \tag{RP-3}$$

- *for all observations $t, k \in T$ and states $s, s' \in S$, if $q_s^t > q_{s'}^k$, then:*

$$\nu_s^t \leq \nu_{s'}^k. \tag{RP-4}$$

The first condition repeats (RP-3) of Theorem 1. The interpretation of ν_s^t as being the derivative (sup-differential) of the function $u(q_s^t)$ is similar to before. The second condition (RP-4) captures the restriction that with a concave function, higher quantities q_s^t lead to lower derivatives. Notice that the values $\widehat{\pi}^t$ can readily be computed from the dataset if the capacity μ is known. For a given capacity μ, conditions (RP-3) and (RP-4) amount to verifying the feasibility of a set of linear inequalities, which can be done efficiently. As mentioned before, we will conduct a grid search on the possible values of μ.

III The SEU model

To understand how well the CMEU model can explain the data, we will compare its performance to the SEU model. Towards this end, we first develop a revealed preference characterization for this model. The revealed preference conditions for SEU rationalization have been established by Echenique and Saito (2015). Here, we give an alternative set of necessary and sufficient conditions that are easier to verify empirically and somewhat more in line with our characterization of CMEU rationalizability of Theorem 2.

Consider a data set $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$. We say that this data is consistent with the subjective expected utility model if, for each state $s \in S$, the DM has a subjective probability π_s (which is independent of the particular observation) that this state will occur and maximizes expected utility given these subjective probabilities.

Definition 2 (SEU rationalizability)**.** *A data set $D = (p^t, q^t, m^t)_{t \in T}$ is subjective expected utility rationalizable if there exists a concave and strictly increasing Bernoulli utility index $u : \mathbb{R}_+ \to \mathbb{R}$ and a probability distribution $\pi \gg 0$ such that for all observations:*[19]

$$q^t \in \arg\max_{q \geq 0} \sum_{s \in S} \pi_s \cdot u(q_s) \ \ s.t. \ \ \sum_s p_s^t \cdot q_s \leq m^t.$$

The following theorem gives a characterization of SEU rationalizability.

[19]The requirement that $\pi \gg 0$ is without loss of generality. If $\pi_s = 0$ for some state s, then utility maximizing behavior will require that $q_s^t = 0$ for all observations $t \leq T$. In such a case, we can omit state s from the analysis.

Theorem 3. *Let $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ be a dataset. Then $\mathcal{D}$ is subjective expected utility rationalizable if and only if there exists a probability distribution $\pi \gg 0$ and for all observations, $t \leq T$ there exists a vector $\nu^t \gg 0$ and a number $\lambda^t > 0$ such that:*

- *for all $t \in T$, $s \in S$*

$$\pi_s \cdot \nu_s^t = \lambda^t \cdot p_s^t \text{ if } q_s^t > 0, \qquad \text{(SEU-1)}$$
$$\pi_s \cdot \nu_s^t \leq \lambda^t \cdot p_s^t \text{ if } q_s^t = 0.$$

- *for all observations $t, k \in T$ and states $s, s' \in S$, if $q_s^t > q_{s'}^k$, then:*

$$\nu_s^t \leq \nu_{s'}^k. \qquad \text{(SEU-2)}$$

Notice the similarity with the conditions in Theorem 2. Condition (SEU-2) is identical to condition (RP-4). Condition (SEU-1) is analogues to condition (RP-3): the only difference is that the probability vectors $\widehat{\pi}^t$ are replaced by the subjective probability vector π.

The restrictions in Theorem 3 are non-linear in the unknowns. In particular SEU-1 contains the product of both π_s and ν_s^t. To resolve this, we perform a grid search on the possible values of π.[20]

IV Implementation and Illustration

We illustrate our results using two experimental data sets from Ahn, Choi, Gale, and Kariv (2014) and Hey and Pace (2014). These data sets have distinct experimental settings, showing that our algorithm is applicable in various situations. Moreover, Hey and Pace (2014) is also instructive as it allows us to assess the performance of the CMEU model outside the typical Ellsberg urn experimental setting.

Before we start, it is helpful to give a short overview of the criteria we will use to assess the performance of the two models. In practice, a data set under consideration will usually not satisfy the (strong) conditions of Theorems 2 or 3. Moreover, as argued by Varian (1990), exact consistency with the underlying decision-making model might not be a very interesting hypothesis. What is more important is to know whether a decision maker's

[20]Notice here that π is independent of the observation $t \leq T$, so the grid size will only depend on the size of the set S.

behavior is close to satisfying the revealed preference conditions. If so, the model still provides a valuable explanation for the observed behavior.

In response to the sharp nature of the deterministic revealed preference tests, it is customary to complement the tests with a goodness-of-fit measure that quantifies how close the observed behavior is to passing the strict revealed preference test. For our application, we will use the goodness-of-fit measure proposed by Houtman and Maks (1985).[21] The Houtman-Maks index (HM) gives the maximal fraction of observations within a given dataset that is still consistent with the revealed preference conditions. For example, an HM-index of 0.5 means that we can (at best) form a sub-dataset containing half the original dataset's observations, satisfying the revealed preference test. We use the HM-index as it is easy to understand and directly applicable to a wide variety of revealed preference models, like the CMEU and SEU models that we consider.[22]

A second issue that may arise when comparing revealed preference tests is that one model performs better, e.g. in terms of the HM-index, simply because it imposes a less strong set of conditions on the observations in the dataset. As such, it is crucial not only to compare the performance between the two models but also to verify whether one of the models is more permissive than the other. We do this by computing the Bronars power index of the revealed preference test (Bronars, 1987). The Bronars index can be computed by generating many random datasets and calculating the fraction of these datasets that violates the particular revealed preference test. A high Bronars index points to a rigorous test, while a low Bronars index indicates that the revealed preference test is weak. Alternatively, one can compute the HM-index for every random dataset. If the obtained distribution of the HM-index from the random sample is (considerably) below that obtained from the actual sample, the test performs rather well. Indeed, in such cases, the HM-indices for the data are higher than those obtained from the simulations. On the other hand, if the sample distribution does not clearly dominate the distribution of the HM-index for the random simulations, the data does not support the model.

A Illustration I: A portfolio choice experiment

Our first illustration uses the dataset gathered by Ahn, Choi, Gale, and Kariv (2014). They ran an experiment where 154 subjects chose a portfolio allocation between two ambiguous

[21] The mixed integer programming problem used to compute the HM-index is presented in Appendix C.

[22] More popular goodness-of-fit measures like the Afriat critical cost-efficiency index (Varian, 1990) only have a clear, intuitive interpretation for particular revealed preference tests like GARP, which looks at rationalizability with the standard utility maximization model.

assets and one risky asset while satisfying a budget constraint. The experiment consisted of 50 independent decision problems. The budget constraints were randomly generated for each decision problem and each subject.

The assets were Arrow securities over three states of the world, which we denote by $S = \{s_1, s_2, s_3\}$. For all observations, t the probability of state 1, the risky state, is known and fixed at $\pi^t(s_1) = \frac{1}{3}$. The ambiguous states 2 and 3, were selected with unknown probabilities that sum up to $\frac{2}{3}$.

Given that s_1 is unambiguous (and by definition then also its complement $\{s_2, s_3\}$) we impose that $\mu(s_1) = \frac{1}{3}$ and $\mu(s_2, s_3) = \frac{2}{3}$. If so, it can be shown that $\mu(s_1, s_2) = \frac{1}{3} + \mu(s_2)$ and $\mu(s_1, s_3) = \frac{1}{3} + \mu(s_3)$.[23] If we set $\alpha = \frac{1}{3} - \mu(s_2)$ and $\beta = \frac{1}{3} - \mu(s_3)$ we get the following parametrization of the capacity μ in terms of the parameters α, β:

$$
\begin{aligned}
\mu(s_1) &= 1/3 \\
\mu(s_2, s_3) &= 2/3 \\
\mu(s_1, s_2) &= 2/3 - \alpha \\
\mu(s_1, s_3) &= 2/3 - \beta \\
\mu(s_2) &= \frac{1}{3} - \alpha \\
\mu(s_3) &= \frac{1}{3} - \beta \\
\mu(s_1, s_2, s_3) &= 1.
\end{aligned}
$$

The first two conditions impose that $\pi(s_1) = 1/3$, so there is no uncertainty with respect to state 1. This captures the fact that subjects know the probability of state 1 in the experiment. One can verify that the capacity μ is supermodular if and only if $-1/3 \leq \alpha, \beta \leq 1/3$ and $\alpha + \beta \geq 0$. By varying the parameters α, β, we can expand or shrink the core of the capacity.

For our test, we first use Algorithm 2 to determine $\widehat{\pi}^t$ for all $t \leq T$. Once these are determined, we can verify CMEU rationalizability by verifying the system of linear inequalities in Theorem 2. As our set of priors $\Delta(\mu)$ varies with α and β, we perform this procedure for each value of the parameters in a grid of possible values. Note that by changing the level of α and β, we are, in effect, doing a grid search on the capacity μ.

[23] For the first, notice that from supermodularity of μ applied to the sets $A = \{s_1\}$ and $B = \{s_2\}$, we get $\mu(s_1, s_2) \geq \mu(s_1) + \mu(s_2) = \frac{1}{3} + \mu(s_2)$ and using $A = \{s_1, s_2\}, B = \{s_2, s_3\}$, we get $\mu(s_1, s_2, s_3) + \mu(s_2) = 1 + \mu(s_2) \geq \mu(s_1, s_2) + \mu(s_2, s_3) = \mu(s_1, s_2) + \frac{2}{3}$, so $\frac{1}{3} + \mu(s_2) \geq \mu(s_1, s_2) \geq \frac{1}{3} + \mu(s_2)$.

For the implementation, we consider 21 equally spaced values for α and 21 equally spaced values of β in the interval $(-1/3, 1/3)$ and select from those combinations (α, β) the ones for which $\alpha + \beta \geq 0$. We then compute the HM-index for each of these possible values. The resulting index then equals the maximal HM-index over all these grid points.

In order to use Theorem 3 to test the SEU model, we need to fix the possible subjective probabilities $(\pi(s_1), \pi(s_2), \pi(s_3))$. Again, we use a grid of possible values. In order not to bias the results, however, we should restrict ourselves to picking only grid points that are also in the grid for the CMEU test.[24] We do this by selecting the 21 values (α, β) in the two-dimensional CMEU grid that satisfy the restriction that $\alpha + \beta = 0$. For these values, the capacity respects additivity: $\mu(s_2) + \mu(s_3) = \mu(s_2, s_3)$, which makes it a probability measure and therefore, the CMEU model reduces to the SEU model model.

Table 1 summarizes the results of our computations. Column two gives the mean HM-index (standard deviation between brackets) over all subjects. The (average) HM-index for the CMEU model is almost equal to the (average) HM-index of the SEU model. The difference is statistically significant but economically negligible: the difference in the HM-index corresponds to an increase of 0.3 observations on average.[25] For 75% of the subjects the HM-indices for the two models are identical.

We also immediately notice that the HM-index for both models is relatively low (20.6% and 20.0%, respectively). One reason may be that the subjects in the experiments are not behaving according to the CMEU or SEU models. Another reason could be that CMEU and SEU revealed preference tests are relatively strong. To evaluate the performance of the two models, it is therefore essential to consider their respective power, especially since SEU is a special case of CMEU. As discussed above, this is usually done by computing their respective Bronars power indices. For each subject, we generate 10 random datasets with the same budget sets as this subject faced, giving 1540 random datasets in total. Column three shows the average HM-index over these simulated random datasets. We see that, on average, the HM-indices of the random datasets are about the same (11.0% and 10.7%).

Next, we compare the HM-index of the 10 random datasets with those of the subjects corresponding to these random draws. The last column of Table 1 gives the average proportion (over all subjects) of these 10 datasets with an HM-index below the one obtained from the actual choice data. We see that 97.7% of the randomly generated datasets have HM-indices below the index obtained from the actual choice data. This measure is the

[24] We thank an anonymous referee for pointing this out to us.

[25] We compute the p-value using a one-sided paired-measure t-test.

Table 1: Summary statistics for fit and power.

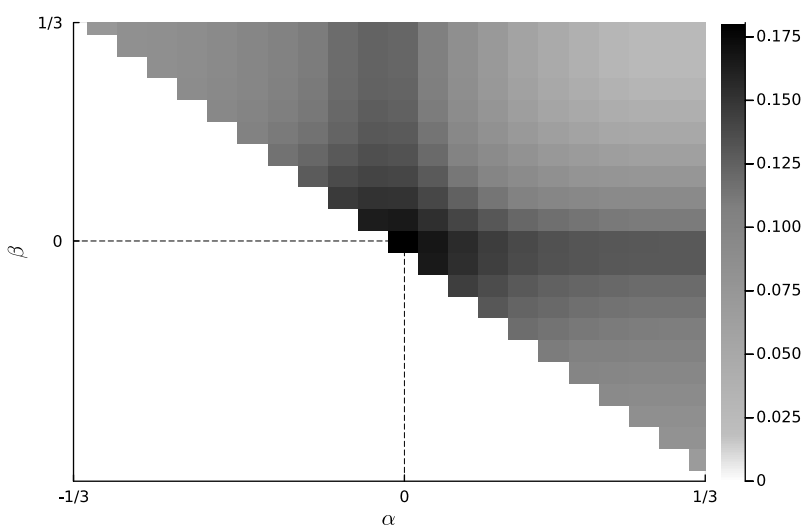
Model	mean HM	mean HM random data	HM random < HM
CMEU	0.206	0.110	0.977
	(0.075)	(0.014)	
SEU	0.200	0.107	0.977
	(0.075)	(0.015)	

Column two gives the mean HM-index over all subjects, standard deviations between brackets. Column 3 gives the mean HM-index over all randomly generated datasets, standard deviations between brackets. The last column gives the fraction of the randomly generated datasets (10 for each subject) with an HM-index below the subjects' HM-index.

same for the two models.

Figure 2 provides more detailed information. The figure gives a heat map for the average (over all subjects) of the HM-index for every value of (α, β) in our grid. We see that the HM-index is (on average) highest for values of α and β close to 0. This coincides with the SEU (CMEU) model where $\pi(s_2) = \pi(s_3) = \frac{1}{3}$. Higher (average) HM-index values can also be observed for supermodular capacities in the neighbourhoods of $\alpha = 0$ and $\beta = 0$.

Figure 2: HM-Index Heatmap



The figure provides a heat map for the average HM-index over all subjects for a given value of (α, β) in the grid.

B Illustration II: The bingo-blower experiment

We use the experimental data set of Hey and Pace (2014) for our second illustration. It is interesting to test our algorithm on this data set as it allows us to compare CMEU's and SEU's performance in an environment that deviates from the standard ambiguity framework with Ellsberg urns. Given the smaller dimension of the dataset, we also test the CMEU model without imposing concavity on the utility index. These results and the theoretical derivations are in Appendix B.

In the experiment, 129 subjects faced a transparent rectangular box filled with fast moving coloured balls: a bingo blower. The bingo blower was filled with 20% pink balls, 30% yellow balls and 50% blue balls. Subjects were asked to bet on the colour of the balls drawn from the bingo blower. They could only guess the proportion of the different colours in the bingo blower because of the fast and erratic movement of the balls. The subjects' inability to precisely determine the proportion of the different colours is the source of the ambiguity in this experiment.

In each round, subjects allocated their tokens between colours. There were two types of problems. Type A problems asked subjects to distribute their tokens between two of the three colours. Type B problems requested them to allocate tokens between one of the colours versus the other two colours. Problems 1 to 3 were of type A, while problems 4 to 6 were of type B. Exchange rates and token endowment changed from one round to another. We split the data set by problem number and apply our algorithm for each of the 6 problems separately. Each problem number was repeated between 9 and 15 times, and each repetition had a different exchange rate and total endowment of tokens.

Similar to the previous illustration, we perform a grid search over the set of possible capacities and select the one that gives the best fit in terms of the HM-index. To exemplify, consider the case where a subject has to bet on *pink* or *yellow*. Since the two states $\{pink, yellow\}$ are the only relevant states, conditional on one of these events happening, we have that the actual values of $\pi(pink)$ and $\pi(yellow)$ are equal to 0.4 and 0.6. Introducing parameters α and β such that $\alpha \in (-0.6, 0.4)$, $\beta \in (-0.4, 0.6)$ and $\alpha + \beta \geq 0$ we can parametrize any supermodular capacity by:

$$\mu(yellow) = 0.6 - \beta$$
$$\mu(pink) = 0.4 - \alpha$$
$$\mu(pink, yellow) = 1.$$

We perform a grid search on both α and β to find the level that gives the best fit in terms of maximizing the HM-index. In particular we consider 11 equally spaced values for α and 11 equally spaced values for β in the intervals $(-0.6, 0.4)$ and $(-0.4, 0.6)$ respectively and retain all combinations (α, β) for which $\alpha + \beta \geq 0$.[26] Similarly, for the SEU model, we proceed as in the previous section and consider only the values (α, β) from the grid that satisfy $\alpha + \beta = 0$.

We summarize the analysis results in Table 2. Rows two and four give the average of the HM-index for the CMEU and SEU models (standard deviations are between brackets). The average values for the HM-index in the CMEU model range between 0.641 and 0.731, while the values for the SEU model range between 0.640 and 0.730, depending on the problem type. The difference of the HM-index for the SEU model is not statistically different from the HM-index of the CMEU model for all problem types.

For our power analysis, we now draw 1000 random datasets.[27] For each random dataset, we compute the fraction of random data sets with an HM-index below the HM-index calculated for the actual subjects. The third and sixth row gives the average fraction of the random datasets with an HM-index below the HM-index of the actual data sets. This proportion is relatively high for both the SEU and CMEU models.

Table 2: Average goodness of fit and Power

		Problem number					
		1	2	3	4	5	6
CMEU	HM-index	0.731	0.641	0.693	0.661	0.697	0.706
		(0.117)	(0.117)	(0.132)	(0.138)	(0.109)	(0.128)
	HM rand < HM	0.971	0.932	0.874	0.638	0.976	0.889
SEU	HM-index	0.730	0.640	0.693	0.661	0.696	0.705
		(0.118)	(0.117)	(0.132)	(0.138)	(0.109)	(0.128)
	HM rand < HM	0.968	0.932	0.874	0.638	0.976	0.888

Lines 2 and 4 give the average HM-index for the CMEU and the SEU models, standard deviations between brackets. Lines 3 and 5 give the average fraction of random datasets with an HM-index below the HM-index of the actual dataset.

Our illustrations indicate that changing the source of the ambiguity does not alter our result: the CMEU and the SEU model seem to perform equally well on both illustrations.

[26] We use this relatively coarse 11-point grid in this illustration to compare the CMEU result with the more computationally demanding non-concave CMEU test that is presented in the Appendix B. Using a denser grid does not alter the qualitative interpretation of the results; see Appendix D.

[27] As the budget sets are identical over all subjects, we do not need to draw random datasets for each subject separately.

V Conclusion

In this paper, we set out an algorithm that provides an efficient way to test for consistency of choice behaviour with the CMEU model of ambiguity aversion. We apply this algorithm using two experimental datasets and compare the fit between the CMEU and the standard SEU model. The CMEU and the SEU model perform equally well in terms of empirical fit in both experiments. Although we cannot comment on the Choquet Expected utility general performance as we focus on its ambiguity averse version, introducing supermodular capacities to model ambiguity aversion does not seem to make a sizeable improvement in terms of model's performance in our two illustrations. As such, if one believes that subjects do exhibit ambiguity aversion on the data sets at hand, then CMEU does not seem to provide a more accurate description of the observed choice behaviour compared to the SEU model.

In this paper, we focused on testing the CMEU model. If a dataset is rationalizable, a follow-up question would be to see whether the primitives of the model, like the utility index, are identified. As usual in revealed preference analysis, point identification is not possible, although it is usually possible to obtain set identification.[28] For example, the conditions in Theorem 2 allow us to obtain bounds on the marginals (slopes) ν_s^t of the utility index (up to normalization). Together with concavity, these could be used to bound the utility index (up to an arbitrary constant and scale).

A Proofs

A Proof of Theorem 1

Assume that the data set is rationalizable, i.e. there exists a monotone and concave u and a convex capacity μ such that for all $t \in T$,

$$q^t \in \arg\max_{q \geq 0} \left\{ \min_{\pi \in \Delta(\mu)} \sum_{s \in S} \pi_s \cdot u(q_s) \right\} \text{ s.t. } \sum_{s \in S} p_s^t \cdot q_s \leq m^t.$$

The objective function $\sum_{s \in S} \pi_s \cdot u(q_s)$ is convex (linear) in π and concave in q, so from the max-min theorem, we know that the optimal solution (π^t, q^t) is a saddle point. This

[28] See Varian (1982) for set identification of such primitives in a general revealed preference setting.

means that for all $q \geq 0$ with $\sum_{s \in S} p_s^t \cdot q_s \leq m^t$ and for all $\pi \in \Delta(\mu)$,

$$\sum_{s \in S} \pi_s^t \cdot u(q_s) \leq \sum_{s \in S} \pi_s^t \cdot u(q_s^t) \leq \sum_{s \in S} \pi_s \cdot u(q_s^t). \tag{2}$$

If we define $u_s^t = u(q_s^t)$ then the second inequality gives immediately that for all $\pi \in \Delta(\mu)$,

$$\sum_{s \in S} \pi_s^t \cdot u_s^t \leq \sum_{s \in S} \pi_s \cdot u_s^t.$$

This gives condition (RP-1). Next, observe that the first inequality of (2) requires:

$$q^t \in \arg\max_{q \geq 0} \sum_{s \in S} \pi_s^t \cdot u(q_s) \text{ s.t. } \sum_{s \in S} p_s^t \cdot q_s \leq m^t.$$

Given that u is concave, the first order conditions for this problem give:

$$\pi_s^t \cdot u'(q_s^t) = \lambda^t \cdot p_s^t \text{ if } q_s^t > 0,$$
$$\pi_s^t \cdot u'(q_s^t) \leq \lambda^t \cdot p_s^t \text{ if } q_s^t = 0,$$

where $\lambda^t > 0$ is the Lagrange multiplier associated with the budget constraint.[29] Concavity of u also implies that for all observations $t, k \in T$ and states $s, s' \in S$:

$$u(q_s^t) - u(q_{s'}^k) \leq u'(q_{s'}^k) \cdot (q_s^t - q_{s'}^k).$$

If we define $u_s^t = u(q_s^t)$, $u_{s'}^k = u(q_{s'}^k)$, $\nu_s^t = u'(q_s^t)$ and $\nu_{s'}^k = u'(q_{s'}^k)$, then we obtain conditions (RP-2) and (RP-3) as we wanted to show.

For the reverse, define the utility index:

$$u(q) = \min_{t \in T, s \in S} u_s^t + \nu_s^t \cdot (q - q_s^t).$$

Let us first show that $u(q_s^t) = u_s^t$. Indeed, first notice that, by definition, $u(q_s^t) \leq u_s^t$. Next, if, towards a contradiction, there is a $k \in T$ and $s' \in S$ such that,

$$u(q_s^t) = u_{s'}^k + \nu_{s'}^k (q_s^t - q_{s'}^k) < u_s^t,$$

we obtain a contradiction with (RP-2).

[29] If u is not differentiable at q_i^t we replace $u'(q_s^t)$ by suitable supgradients.

It is also easily verified that u is monotone (as $\nu_s^t \gg 0$) and concave (as it is the minimum of a set of linear functions). Let us now show that $u(.)$ CMEU rationalizes the dataset $\mathcal{D}$. Towards a contradiction, assume that there is a q such that for some $t \in T$, $\sum_{s\in S} p_s^t \cdot q_s \leq m^t$ and,

$$\min_{\pi\in\Delta(\mu)} \sum_{s\in S} \pi_s \cdot u(q_s) > \min_{\pi\in\Delta(\mu)} \sum_{s\in S} \pi_s \cdot u_s^t.$$

From (RP-1), π^t solves the minimization problem on the right-hand side of the inequality. This gives,

$$\min_{\pi\in\Delta(\mu)} \sum_{s\in S} \pi_s \cdot u(q_s) > \sum_{s\in S} \pi_s^t \cdot u_s^t.$$

Let π^* solve the minimization problem on the left-hand side. Then,

$$\begin{aligned}
\sum_{s\in S} \pi_s^* \cdot u(q_s) &\leq \sum_{s\in S} \pi_s^t \cdot u(q_s), \\
&\leq \sum_{s\in S} \pi_s^t \cdot u_s^t + \sum_{s\in S} \pi_s^t \nu_s^t \cdot (q_s - q_s^t), \\
&\leq \sum_{s\in S} \pi_s^t \cdot u_s^t + \lambda^t \sum_{s\in S} p_s^t \cdot (q_s - q_s^t), \\
&\leq \sum_{s\in S} \pi_s^t \cdot u_s^t,
\end{aligned}$$

The first inequality is obtained from (RP-1). The second inequality follows from the definition of $u(.)$. The third inequality is a consequence of (RP-3), and finally, the last inequality follows from the fact that $\sum_{s\in S} p_s^t \cdot q_s \leq m_t = \sum_{s\in S} p_s^t \cdot q_s^t$. This gives the desired contradiction.

B Proof of Proposition 1

Let $\mathcal{D} = (p^t, q^t, m^t)_{t\leq T}$ be a dataset that satisfies (RP-1)-(RP-3). Let $(\pi^t, u^t, \nu^t, \lambda^t)_{t\leq T}$ be a solution to (RP-1)-(RP-3). Then according to (RP-1):

$$\pi^t \in \arg\min \sum_{s\in S} \pi_s \cdot u_s^t \text{ subject to } \pi \in \nabla(V).$$

As $\nabla(V)$ is a polymatroid, π^t can be generated as the output of Algorithm 1. Let $I_1^t, \ldots, I_K^t$ be the equivalence classes defined in the main text. Notice that for $j < i$, all states in I_j^t are

processed before I_i^t in Algorithm 1. Let $I_i^t = \{s_{(k+1)}, s_{(k+2)}, \dots s_{(k+|I_i^t|)}\}$ where we rank the states according to the iteration in Algorithm 1 that generates π^t. Then, for Algorithm 1:

$$\pi^t_{s_{(k+j)}} = V(L_i^t \cup \{s_{(k+1)}, \dots s_{(k+j)}\} - V(L_i^t \cup \{s_{(k)} \dots, s_{(k+j-1)}\}.$$

Adding up over $j = 1, \dots, |I_j^t|$ gives:

$$\sum_{s \in I_i^t} \pi_s^t = V(L_i^t \cup I_i^t) - V(L_i^t) = \widetilde{V}_i^t(I_i^t). \tag{3}$$

This shows (C.2). For (C.1), let $A \subseteq I_i^t$. Notice that as $\pi^t \in \nabla(V)$:

$$\sum_{s \in L_i^t} \pi_s^t + \sum_{s \in A} \pi_s^t \leq V(L_i^t \cup A).$$

Also from (3), we have for the first term:

$$\sum_{s \in L_j^t} \pi_s^t = \sum_{j<i} \sum_{s \in I_j^t} \pi_s^t = \sum_{j<i} \left(V(I_j^t \cup L_i^t) - V(L_i^t) \right).$$

As $L_{i+1}^t = L_i^t \cup I_i^t$, this gives:

$$\sum_{s \in L_i^t} \pi_s^t = V(L_i^t).$$

As such:

$$\begin{aligned} V(L_i^t) + \sum_{s \in A} \pi_s^t &= \sum_{s \in L_i^t} \pi_s^t + \sum_{s \in A} \pi_s^t \leq V(L_i^t \cup A), \\ \rightarrow \sum_{s \in A} \pi_s^t &\leq V(A \cup L_i^t) - V(L_i^t) = \widetilde{V}_i^t(A), \end{aligned}$$

as was to be shown.

For the second part, assume that $(\pi^t)_{t \leq T}$ satisfy (C.1)-(C.2). Let us first show that it satisfies (RP-1). Let π solve:

$$\pi \in \arg\min \sum_{s \in S} \pi_s \cdot u_s^t \text{ s.t. } \pi \in \nabla(V).$$

As V is submodular, this is a polymatroid minimization problem. As such, its solution can be obtained using Algorithm 1. As in the first part of the proof, we can show that for all

I_i^t:

$$\sum_{s\in I_i^t}\pi_s = V(L_i^t\cup I_i^t) - V(L_i^t) = \sum_{s\in I_i^t}\pi_s^t,$$

where the last equality follows from (C.2). For $s\in I_i^t$, let $u_s^t = U_i^t$, which indeed only depends on i. Then:

$$\begin{aligned}\sum_{s\in S}\pi_s\cdot u_s^t &= \sum_i\sum_{s\in I_i^t}\pi_s\cdot u_s^t,\\ &= \sum_i U_i^t\sum_{s\in I_i^t}\pi_s,\\ &= \sum_i U_i^t\sum_{s\in I_i^t}\pi_s^t,\\ &= \sum_{s\in S}\pi_s^t\cdot u_s^t.\end{aligned}$$

This shows that π_s^t also solves (RP-1).

Next we need to show that $\pi^t\in\nabla(V)$, i.e. for all $A\subseteq S$, $\sum_{s\in A}\pi_s^t\le V(A)$. Define:

$$A_i = A\cap I_i^t \text{ with } A_0=\emptyset.$$

Notice that $L_i^t\cup A_i = L_i^t\bigcup(\cup_{j\le i}A_j)$, and that $L_i^t\bigcap(\cup_{j\le i}A_j) = \cup_{j<i}A_j$. As such, using the submodularity of V:

$$\begin{aligned}V(L_i^t\cup A_i) &= V\left(L_i^t\cup(\cup_{j\le i}A_j)\right),\\ &\le V(L_i^t) + V(\cup_{j\le i}A_j) - V(\cup_{j<i}A_j).\end{aligned}$$

As such, using (C.2):

$$\begin{aligned}\sum_{s\in A}\pi_s^t &= \sum_i\sum_{s\in A_i}\pi_s^t,\\ &\le \sum_i \widetilde{V}_i^t(A_i),\\ &= \sum_i\left(V(L_i^t\cup A_i) - V(L_i^t)\right),\\ &\le \sum_i\left(V(\cup_{j\le i}A_j) - V(\cup_{j<i}A_j)\right),\\ &= V(\cup_j A_j) = V(A),\end{aligned}$$

as was to be shown.

C Proof of Proposition 2

First of all, notice that $\varepsilon_s^m = 0$ for all $s \in I_i^t$ is feasible at each optimization problem. Also, the constraint set is compact and closed. As such, the optimization problem is well-defined.

Fact 1: The set A^m in step 4 of Algorithm 2 is well defined and unique.

To see this, let A and B be subsets of I_i^t for which (Lm.4) is binding. Then from submodularity of the function $\widetilde{V}_i^t$ we get:

$$\begin{aligned}\sum_{s\in A}\sum_{j=0}^{m}\varepsilon_s^j + \sum_{s\in B}\sum_{j=0}^{m}\varepsilon_s^j = &\sum_{s\in A\cup B}\sum_{j=0}^{m}\varepsilon_s^j + \sum_{s\in A\cap B}\sum_{j=0}^{m}\varepsilon_s^j,\\ \leq &\widetilde{V}_i^t(A\cup B) + \widetilde{V}_i^t(A\cap B),\\ \leq &\widetilde{V}_i^t(A) + \widetilde{V}_i^t(B),\\ = &\sum_{s\in A}\sum_{j=0}^{m}\varepsilon_s^j + \sum_{s\in B}\sum_{j=0}^{m}\varepsilon_s^j\end{aligned}$$

So all inequalities are equalities, which means that (Lm.4) is also binding for $A \cup B$.

Fact 2: For all m, $A^m \subsetneq A^{m+1}$.

That $A^m \subseteq A^{m+1}$ holds from the fact that condition (L(m+1).4) is binding for A^m (as $\varepsilon_s^{m+1} = 0$ for all $s \in A^m$ and the restriction (Lm.4) is binding for A^m).

Towards a contradiction, assume that $A^m = A^{m+1}$ with optimal values ε_s^{m+1}.

Now, from the maximality and uniqueness of $A^{m+1} = A^m$ we have that for all sets B with $B \cap \Omega^{m+1} \neq \emptyset$, condition (L(m+1).4) is slack, i.e.

$$\sum_{s\in B}\sum_{j=0}^{m+1}\varepsilon_s^j < \widetilde{V}_i^t(B).$$

(Otherwise, A^{m+1} would contain elements in Ω^{m+1} which is disjoint from A^m by definition). Let $\delta > 0$ be a small enough value such that if we set for all $s \in \Omega^{m+1}$,

$\widetilde{\varepsilon}_s^{n+1} = \varepsilon_s^{m+1} + \delta p_s^t$, then for all these B,

$$\sum_{s\in B}\left(\sum_{j=0}^{m}\varepsilon_s^j + \widetilde{\varepsilon}_s^{m+1}\right) \leq \widetilde{V}_i^t(B).$$

(For $s \notin \Omega^{m+1}$ we set $\widetilde{\varepsilon}_s^{m+1} = 0$). Notice that these values of $\widetilde{\varepsilon}_s^{m+1}$ satisfy the conditions $(L(m+1).1)$, $(L(m+1).2)$, $(L(m+1).3)$ and $(L(m+1).4)$. Also, the objective function for this feasible value gives

$$\sum_{s\in\Omega^{m+1}}\widetilde{\varepsilon}_s^{m+1} = \sum_{s\in\Omega^{m+1}}\varepsilon_s^{m+1} + \sum_{s\in\Omega^{m+1}}\delta p_s^t > \sum_{s\in\Omega^{m+1}}\varepsilon_s^{m+1}$$

a contradiction with the optimality of ε_s^{m+1}.

Fact 3: $\sum_{s\in I_i^t}\pi_s^t = \widetilde{V}_i^t(I_i^t)$.

Indeed, the previous item shows that the algorithm ends in finite time, as at some value $m = M$ where $A^M = I_i^t$ and then $\Omega^{M+1} = \emptyset$. Then it follows that

$$\sum_{s\in I_i^t}\pi_s^t = \sum_{s\in I_i^t}\sum_{j=0}^{M}\varepsilon_s^j = \widetilde{V}_i^t(A^M) = \widetilde{V}_i^t(I_i^t).$$

To show that $\widehat{\pi}^t$ solves both **OP-I** and **OP-II**, let

$$LB = \min_{s\in I_i^t}\frac{\widehat{\pi}_s^t}{p_s^t}.$$

Notice that this minimum is reached for a state $s \in A^0$ with value $LB = \frac{\varepsilon_s^0}{p_s^t}$ for any $s \in A^0$

We will show that this is the max-min value over all probability distributions. Assume not, then there is a probability distribution $\pi_{I_i^t}$ that satisfies the constraints of **OP-I** and such that

$$\min_{s\in I_i^t}\frac{\pi_s}{p_s^t} > \min_{s\in I_i^t}\frac{\widehat{\pi}_s^t}{p_s^t}.$$

In particular, this means that for all $s \in A^0$, $\pi_s > \widehat{\pi}_s^t$. But then:

$$\widetilde{V}_i^t(A^0) = \sum_{s\in A^0}\varepsilon_s^0 = \sum_{s\in A^0}\widehat{\pi}_s^t < \sum_{s\in A^0}\pi_s \leq \widetilde{V}_i^t(A^0),$$

A contradiction. The first inequality follows from the fact that for A^0, condition (L0.4) is

binding. Next, let

$$UB = \max_{s \in I_i^t} \frac{\widehat{\pi}_s^t}{p_s^t}.$$

Notice that this maximum is reached for a state $s \in \Omega^M$ (where M is the last iteration of the Algorithm). In particular, the value of UB is equal to $\frac{\sum_{j=0}^{M} \varepsilon_s^j}{p_s^t} = \frac{\widehat{\pi}_s^t}{p_s^t}$ for all $s \in \Omega^M$.

We will show that UB is the min-max value. Towards a contradiction, assume that there is a probability distribution $\pi_{I_i^t}$ that satisfies the constraints of **OP-II** such that,

$$\max_{s \in I_i^t} \frac{\pi_s}{p_s^t} < \max_{s \in I_i^t} \frac{\widehat{\pi}_s^t}{p_s^t}.$$

In particular, this means that for all $s \in \Omega^M$, we have that $\pi_s < \widehat{\pi}_s^t$. But then:

$$\begin{aligned}
&\sum_{s \in \Omega^M} \pi_s < \sum_{s \in \Omega^N} \widehat{\pi}_s^t, \\
\leftrightarrow &\widetilde{V}_i^t(I_i^t) - \sum_{s \in \Omega^M} \pi_s > \widetilde{V}_i^t(I_i^t) - \sum_{s \in \Omega^M} \widehat{\pi}_s^t, \\
\leftrightarrow &\sum_{s \in A^{M-1}} \pi_s > \sum_{s \in A^{M-1}} \widehat{\pi}_s^t = \sum_{s \in A^{M-1}} \sum_{j=0}^{M-1} \varepsilon_s^j = \widetilde{V}(A^{M-1})
\end{aligned}$$

Here we made use of the fact that for any probability distribution $\sum_{s \in I_i^t} \pi_s = \widetilde{V}_i^t(I_i^t)$. This gives the desired contradiction.

D Proof of Corollary 1

Let $\mathcal{D}$ be CMEU rationalizable and let $(\pi^t, u^t, \nu^t, \lambda^t)$ be a set of feasible values of (RP-1)–(RP-3). If I_i^t is a singleton, say $\{s\}$, then by proposition 1 and the discussion in the main text, $\widehat{\pi}_s^t = \pi_s^t$, as such it satisfies (RP-2)–(RP-3). If there is more than one state in I_i^t, by the reasoning outlined above 2, replacing π_s^t by $\widehat{\pi}_s^t$ for all $s \in I_i^t$ would also satisfy (RP-2)–(RP-3). Sufficiency is obvious given Theorem 1 and the second part of Proposition 1.

E Proof of Theorem 2

From Proposition 2, we know that to verify whether a dataset $\mathcal{D}$ is CMEU rationalizable, we only need to verify that there exist vectors u^t and $\nu^t \gg 0$ and numbers λ^t such that $(\widehat{\pi}^t, u^t, \nu^t, \lambda^t)$ satisfies conditions (RP-2) and (RP-3), where $\widehat{\pi}^t$ is defined by the output of Algorithm 2.

As such, we are left to show that there exist vectors u^t and $\nu^t \gg 0$ and numbers λ^t satisfying (RP-2) and (RP-3) if and only if there exist vectors $\nu^t \gg 0$ and numbers $\lambda^t > 0$ satisfying (RP-3) and (RP-4).

Assume that (RP-2) and (RP-3) are satisfied. Then obviously (RP-3) is also satisfied. Next, for all observations $t, k \leq T$ and states $s, s' \in S$:

$$u_s^t - u_{s'}^k \leq \nu_{s'}^k \cdot (q_s^t - q_{s'}^k),$$
$$u_{s'}^k - u_s^t \leq \nu_s^t \cdot (q_{s'}^k - q_s^t).$$

Adding these two inequalities together gives:

$$0 \leq (\nu_{s'}^k - \nu_s^t) \cdot (q_s^t - q_{s'}^k).$$

As such,

$$q_s^t > q_{s'}^k \text{ implies } \nu_{s'}^k \geq \nu_s^t.$$

which gives (RP-4)

Next, let us show that (RP-3) and (RP-4) imply (RP-2).

Partition all the values of q_s^t into equivalence classes $E_1, ..., E_N$ such that $\forall t, k \leq T$ and $s, s' \in S$, $q_s^t, q_{s'}^k \in E_i \leftrightarrow q_s^t = q_{s'}^k$. Let $q[i]$ be a representative member of the equivalence class E_i. We index the equivalence classes such that $q[i] < q[j]$ if and only if $i < j$. Now assign to every equivalence class E_i values $\nu^-[i]$ and $\nu^+[i]$ such that: $\forall i \in \{1, ..., N\}$:

$$\nu^-[i] = \min_{q_s^t \in E_1} \nu_s^t \qquad \text{and} \qquad \nu^+[i] = \max_{q_s^t \in E_i} \nu_s^t.$$

Defined this way, $\nu^-[i] \leq \nu_s^t \leq \nu^+[i]$ for all $q_s^t \in E_i$ and $\nu^+[i+1] \leq \nu^-[i]$ (by (RP-4)). Next, consider the linear interpolation $f : \mathbb{R}_+ \to \mathbb{R}_{++}$

$$f(q) = \begin{cases} \nu^+[1] & \text{if } q < q[1], \\ \nu^-[i] + \frac{\nu^+[i+1] - \nu^-[i]}{q[i+1] - q[i]}(q - q[i]) & \text{if } q \in [q[i], q[i+1]) \\ \nu^-[N] & \text{i f} q \geq q[N]. \end{cases}$$

This is a non-increasing function. Furthermore, for all i, $f(q[i]) = \nu^-[i]$ and $\lim_{\substack{q \to q[i+1] \\ q < q[i+1]}} f(q) = \nu^+[i+1]$, so f is a càdlàg function (continuous from the right with left limits).

Now define the function

$$u(q) = \int_0^q f(z)dz$$

Since $f(q) > 0$ and non-increasing, we have that the function $u(q)$ is increasing, continuous and concave. In particular for all $q[i]$, we have that:

$$\partial u(q[i]) = [\nu^-[i], \nu^+[i]].$$

where $\partial u(q)$ is the set of sup-differentials of u (Rockafellar (1970)). In particular, for all $t \leq T$ and $s \in S$, $\nu_s^t \in \partial u(q_s^t)$ Since u is concave, it follows from the definition of a sup-differential that for all observations t, k and states $s, s' \in S$:

$$u(q_s^t) - u(q_{s'}^k) \leq \nu_{s'}^k \cdot (q_s^t - q_{s'}^k) \tag{4}$$

If we define $u_s^t \equiv u(q_s^t)$, then (4) gives (RP-2).

F Proof of Theorem 3

Assume that the dataset $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ is SEU rationalizable and let $u : \mathbb{R} \to \mathbb{R}$ be the monotone concave Bernoulli utility index. By the first order conditions:

$$q_s^t > 0 \to u'(q_s^t) = \lambda^t \cdot \frac{p_s^t}{\pi_s},$$
$$q_s^t = 0 \to u'(q_s^t) \leq \lambda^t \cdot \frac{p_s^t}{\pi_s}.$$

Here $\lambda^t > 0$, is the Lagrange multiplier associated with the budget constraint. By replacing $u'(q_s^t)$ by ν_s^t, we get (SEU-1). Now given that u is concave, we have that for all $t, k \leq T$ and $s, s' \in S$,

$$u(q_s^t) - u(q_{s'}^k) \leq \nu_{s'}^k \cdot (q_s^t - q_{s'}^k),$$
$$u(q_{s'}^k) - u(q_s^t) \leq \nu_s^t \cdot (q_{s'}^k - q_s^t).$$

Adding these two inequalities together gives:

$$0 \leq (\nu_{s'}^k - \nu_s^t) \cdot (q_s^t - q_{s'}^k).$$

As such,

$$q_s^t > q_{s'}^k \text{ implies } \nu_{s'}^k \geq \nu_s^t.$$

which gives (SEU-2).

Now, for the reverse, assume that (SEU-1) and (SEU-2) are satisfied. Construct a function $u : \mathbb{R}_+^n \to \mathbb{R}$ as in the proof of Theorem 2. The function u is strictly increasing, continuous and concave and for all $t \in T$ and $s \in S$, $\nu_s^t \in \partial u(q_s^t)$.

Let us show that u rationalizes the data set given the probabilities π. Towards a contradiction assume there is a bundle q such that $\sum_{s\in S} \pi_s \cdot u(q_s) > \sum_{s\in S} \pi_s \cdot u(q_s^t)$ and $\sum_{s\in S} p_s^t \cdot q_s \leq m^t$.

Then:

$$\begin{aligned}
0 < \sum_{s\in S} \pi_s \cdot u(q_s) - \sum_{s\in S} \pi_s \cdot u(q_s^t) &= \sum_{s\in S} \pi_s \cdot (u(q_s) - u(q_s^t)), \\
&\leq \sum_{s\in S} \pi_s \nu_s^t \cdot (q_s - q_s^t), \\
&\leq \sum_{s\in S} \lambda^t p_s^t \cdot (q_s - q_s^t), \\
&= \lambda^t \sum_{s\in S} p_s^t \cdot (q_s - q_s^t) \leq 0,
\end{aligned}$$

a contradiction. The first inequality follows from the fact that u is concave, and the second from (SEU-2).

G Illustration

Let us investigate the performance of the CMEU model on the portfolio choice experiment. Notice that as before, $\pi(s_1) = 1/3$. Next we need to show how we distribute the probability $\mu(s_2, s_3) = \frac{2}{3}$ to state s_2 and s_3. When $q^t(s_2) \neq q^t(s_3)$, we can use C.1 and C.2 as in the main text.

If $q^t(s_2) = q^t(s_3)$, we use the algorithm in the proof of lemma 1 to construct $\pi^t(I_i^t)$, where $s_2, s_3 \in I_i^t$. To do this, let us define $V(E) = 1 - \mu(S \setminus E)$. If the algorithm finishes

with $\Omega^1 = \emptyset$. Then we have: $\frac{\pi^t(s_2}{p^t(s_2)} = \frac{\pi^t(s_3}{p^t(s_3)}$ and

$$\pi^t(s_2) = \widetilde{V}(s_2, s_3)\frac{p^t(s_2)}{p^t(s_2) + p^t(s_3)}$$
$$\pi^t(s_3) = \widetilde{V}(s_2, s_3)\frac{p^t(s_3)}{p^t(s_2) + p^t(s_3)}$$

If $\Omega^1 = s_2$, and $\widetilde{V}(s_2, s_3)\frac{p^t(s_3)}{p^t(s_2)+p^t(s_3)} > \widetilde{V}(s_3)$. This means that there is an A, $0 < A < \widetilde{V}(s_2, s_3)$ such that

$$\epsilon^0(s_2) = A * \frac{p^t(s_2)}{p^t(s_2) + p^t(s_3)}$$
$$\epsilon^0(s_3) = A * \frac{p^t(s_3)}{p^t(s_2) + p^t(s_3)} = \widetilde{V}(s_3)$$

Next we must have $\Omega^2 = \emptyset$, and

$$\widetilde{V}(s_2, s_3) = \epsilon^0(s_2) + \epsilon^1(s_2) + \epsilon^0(s_3)$$
$$\iff \pi^t(s_2) = \widetilde{V}(s_2, s_3) - \widetilde{V}(s_3)$$

And similarly if $\Omega^1 = s_3$, we have

$$\pi^t(s_3) = \widetilde{V}(s_2, s_3) - \widetilde{V}(s_2)$$

Using the parametrization of the core $\Delta(\mu)$ described in the main text, we have that if $q^t(s_2) = q^t(s_3)$:

1. If $\frac{p^t(s_2)}{p^t(s_2)+p^t(s_3)}\frac{2}{3} \leq \frac{1}{3} + \alpha$ and $\frac{p^t(s_3)}{p^t(s_2)+p^t(s_3)}\frac{2}{3} \leq \frac{1}{3} + \beta$, then:

$$\pi^t(s_2) = \widetilde{V}(s_2, s_3)\frac{p^t(s_2)}{p^t(s_2) + p^t(s_3)}, \quad \pi^t(s_3) = \widetilde{V}(s_2, s_3)\frac{p^t(s_3)}{p^t(s_2) + p^t(s_3)}.$$

2. If $\frac{p^t(s_2)}{p^t(s_2)+p^t(s_3)}\frac{2}{3} > \frac{1}{3} + \alpha$:

$$\pi^t(s_2) = \frac{1}{3} + \alpha, \quad \pi^t(s_3) = \frac{1}{3} - \alpha.$$

3. If $\frac{p^t(s_3)}{p^t(s_2)+p^t(s_3)}\frac{2}{3} > \frac{1}{3} + \beta$:

$$\pi^t(s_2) = \frac{1}{3} - \beta, \quad \pi^t(s_3) = \frac{1}{3} + \beta.$$

B Non-concave CMEU

In this appendix, we derive the testable implications for rationalizability with CMEU without restricting the utility index to be concave.

Definition 3. *A data set $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ is non-concave CMEU rationalizable for a capacity μ if there exists a strictly increasing Bernoulli utility index $u : \mathbb{R}_+ \to \mathbb{R}$, such that for all observations $t \leq T$,*

$$q^t \in \arg\max_{q \geq 0} \left\{ \min_{\pi \in \Delta(\mu)} \sum_{s \in S} \pi_s \cdot u\big(q_s\big) \right\} \ \ s.t. \ \sum_{s \in S} p^t_s \cdot q_s \leq m^t.$$

Consider the set $\mathcal{X}$ of all the consumption levels that are chosen at some observation in some state together with the zero consumption value, $\mathcal{X} = \{0\} \cup \{q^t_s : t \leq T, s \in S\}$. Defined the grid $\mathcal{G}$ by $\mathcal{G} = \mathcal{X}^n = \mathcal{X} \times \ldots \times \mathcal{X}$. Recall that n is the number of states.

The following is a simple adaptation of a characterization of the revealed preference conditions for the MEU model by Polisson, Quah, and Renou (2020).

Proposition 3 (Adapted from Polisson, Quah, and Renou (2020))**.** *A data set $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ is non-concave CMEU rationalizable for a capacity μ if and only if there is a function $\Phi : \mathcal{G} \to \Delta(\mu)$ and a strictly increasing function $u : \mathcal{X} \to \mathbb{R}_+$ such that:*

- *For all $q \in \left\{q \in \mathcal{G} : \sum_{s \in S} p^t_s \cdot q^t_s \leq m^t\right\}$,*

$$\sum_{s \in S} \Phi(q^t)_s \cdot u(q^t_s) \geq \sum_{s \in S} \Phi(q)_s \cdot u(q_s) \tag{NC-CMEU-1}$$

- *For all $q \in \left\{q \in \mathcal{G} : \sum_{s \in S} p^t_s \cdot q_s < \sum_{s \in S} p^t_s \cdot q^t_s\right\}$:*

$$\sum_{s \in S} \Phi(q^t)_s \cdot u(q^t_s) > \sum_{s \in S} \Phi(q)_s \cdot u(q_s) \tag{NC-CMEU-2}$$

- *For all $q \in \mathcal{G}$ and all $\pi \in \Delta(\mu)$:*

$$\sum_{s \in S} \Phi(q)_s \cdot u(q_s) \leq \sum_{s \in S} \pi_s \cdot u(q_s). \tag{NC-CMEU-3}$$

Condition (NC-CMEU-1) states that the utility obtained from q^t should give at least as much utility as any feasible bundle in the grid $\mathcal{G}$ in the budget set. Condition (NC-CMEU-2) is the strict version of this condition. Finally, condition (NC-CMEU-3) follows from the idea that for all bundles $q \in \mathcal{G}$, the probabilities $\Phi(q)$ minimize expected utility over $\Delta(\mu)$.[30]

The Proposition gives a collection of inequalities to be verified in terms of finitely many unknowns (i.e. the values of $u(q)$ for $q \in \mathcal{X}$ and the values of $\Phi(q)$ for $q \in \mathcal{G}$). However, the inequalities are quadratic (nonlinear) in these unknowns.

We will show that if the capacity μ is known, then we can use our algorithm to recover the functions Φ and transform this system of inequalities into a finite system of linear inequalities. Let us consider the first step of our algorithm, where ties in step 1 of Algorithm 1 are broken in an arbitrary but consistent way. Then our algorithm can give, for any arbitrary bundle $q \in \mathcal{G}$, a unique probability distribution in $\Delta(\mu)$, say $\widehat{\Phi}(q)$ that minimizes $\sum_{\pi \in \Delta(\mu)} \pi_s \cdot u(q_s)$ for any strictly increasing function $u(.)$.

Proposition 4. *Consider a data set $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ and let $\widehat{\Phi}(q)$ be the output of Algorithm 1 for the bundle $q \in \mathcal{G}$.*

Then $\mathcal{D}$ is non-concave CMEU rationalizable for a capacity μ if and only if there is a strictly increasing function $u : \mathcal{X} \to \mathbb{R}^+$ such that:

- *For all $q \in \left\{q \in \mathcal{G} : \sum_{s \in S} p_s^t \cdot q_s \leq \sum_{s \in S} p_s^t \cdot q_s^t\right\}$:*

$$\sum_{s \in S} \widehat{\Phi}(q^t)_s \cdot u(q_s^t) \geq \sum_{s \in S} \widehat{\Phi}(q)_s \cdot u(q_s) \tag{CN-CMEU-4}$$

- *For all $q \in \left\{q \in \mathcal{G} \middle| \sum_{s \in S} p_s^t \cdot q_s < \sum_{s \in S} p_s^t \cdot q_s^t\right\}$:*

$$\sum_{s \in S} \widehat{\Phi}(q^t)_s \cdot u(q_s^t) > \sum_{s \in S} \widehat{\Phi}(q)_s \cdot u(q_s) \tag{CN-CMEU-5}$$

Proof. Let $\mathcal{D}$ be non-concave CMEU rationalizable. By condition (NC-CMEU-3) of Proposition 3, we have that for every $q \in \mathcal{G}$, $\Phi(q)$ is a solution to $\min_{\pi \in \Delta(\mu)} \sum_{s \in S} \pi_s \cdot u\big(q_s\big)$.

[30] This latter condition deviates from the one in Polisson, Quah, and Renou (2020) as they do not assume that the set of priors for the MEU model is known.

According to the first part of the proof of Proposition 1, the distribution $\Phi(q)$ should therefore satisfy (C.1) and (C.2).

Now, let us consider the set of equivalence classes $I_1, \ldots, I_K$ generated by $q \in \mathcal{G}$. If I_i is a singleton, then for $s \in I_i$, we $\Phi(q)_s = \widehat{\Phi}(q)_s$ as both minimize expected utility. If there is more than one state s in I_j, observe that as both $\Phi(q)$ and $\widehat{\Phi}(q)$ satisfy (C.2),

$$\begin{aligned}
\sum_{s \in I_i} \widehat{\Phi}(q)_s \cdot u(q_s) &= \widetilde{V}_i(I_i) \cdot u(q_{s'}) \quad \text{for some } s' \in I_i \\
&= \left(\sum_{s \in I_i} \widehat{\Phi}(q)_s \right) u(q_{s'}) \quad \text{for some } s' \in I_i \\
&= \sum_{s \in I_i} \Phi(q)_s \cdot u(q_s)
\end{aligned}$$

As such, if we substitute $\Phi(q)$ by $\widehat{\Phi}(q)$ in (NC-CMEU-1) and (NC-CMEU-2), the conditions remain the same. This gives (CN-CMEU-4) and (CN-CMEU-5).

For the reverse, assume (CN-CMEU-4) and (CN-CMEU-5) are satisfied. It is immediate that the choice of $\Phi(q) = \widehat{\Phi}(q)$ satisfies (NC-CMEU-1) and (NC-CMEU-2). Next, since (from Algorithm 1) the probabilities $\widehat{\Phi}(q)$ solve $\min_{\pi \in \Delta(\mu)} \sum_{s \in S} \pi_s \cdot u(q_s)$ for any strictly increasing function $u(.)$, we see that (NC-CMEU-3) is also satisfied.

□

A Illustration

The set $\mathcal{X}$ is of the order Tn which means that the set $\mathcal{G}$ is of the order $(Tn)^n$, which is exponential in the number of states. For each bundle $q \in \mathcal{G}$ in this grid, the value $\widehat{\Phi}(q)$ needs to be computed. Next, there is the issue that the potential number of inequalities in (CN-CMEU-4) and (CN-CMEU-5) grows with the size of $\mathcal{G}$. Finally, we have the grid over the possible capacities μ.

Given this, the problem is only computationally feasible for datasets with a limited number of states and observations. We apply the test to the dataset of Hey and Pace (2014) (illustration 2 in the main part of the paper). We use the same grid specification as in the main text. For the non-concave SEU specification, we again select all the points on the grid where $\alpha + \beta = 0$.

Table 3 gives the results. The non-concave CMEU (SEU) model has a higher HM-index than the concave CMEU (SEU) model. This is not surprising given that the non-concave

Table 3: Average goodness of fit and Power for non-concave CMEU

		Problem number					
		1	2	3	4	5	6
non-concave CMEU	HM-index	0.836	0.830	0.839	0.817	0.821	0.804
		(0.093)	(0.091)	(0.094)	(0.102)	(0.100)	(0.095)
	HM rand < HM	0.894	0.881	0.800	0.529	0.902	0.761
non-concave SEU	HM-index	0.824	0.820	0.828	0.810	0.811	0.795
		(0.094)	(0.090)	(0.089)	(0.102)	(0.100)	(0.095)
	HM rand < HM	0.889	0.884	0.813	0.528	0.907	0.768

The first line gives the average HM-index for the non-concave CMEU. The second line gives the standard deviations. The last line gives the average fraction of random datasets with an HM-index below the HM-index of the actual dataset.

models are weaker compared to their concave counterparts. Reversely, we have that the power of the non-concave models will be below the ones of the concave counterparts. The third row gives the proportion of these randomly generated datasets below the actual data's HM-index. These figures are consistently slightly below the ones for the concave CMEU and SEU counterparts. In all, relaxing concavity for the CMEU model does not really seem to improve its empirical fit, for the data under consideration.

If we compare the non-concave CMEU model with the non-concave SEU model, we see the SEU model has a slightly lower fit than the non-concave CMEU which is unsurprising given that the CMEU model contains the SEU model as a special case. The difference between the two HM-indices, however, is qualitatively negligible. The last row indicates that the higher non-concave CMEU HM-index is counterbalanced by a lower power relative to the non-concave SEU model. In all, we find that, also for the non-concave part, the two models seem to perform equally well.

C Mixed integer programming formulation for the HM-index

Let $\mathcal{D} = (p^t, q^t, m^t)_{t \leq T}$ be a dataset and let $\xi = (\xi^1, \ldots, \xi^T)$ be a vector such that $\xi^t \in \{0, 1\}$ for all $t = 1, \ldots, T$.

To compute the HM-index, we need to find the largest set $W \subseteq \{1, \ldots T\}$ such that the dataset $(p^t, q^t, m^t)_{t \in W}$ is CMEU-rationalizable. The idea behind the vector ξ is that $\xi^t = 1$ whenever $t \in W$ and $\xi^t = 0$ if $t \notin W$.

The HM-index can then be found by solving the following mixed integer programming

problem (for unknown $\kappa_s^t, \gamma^t \in \mathbb{R}$ and $\xi^t \in \{0,1\}$):[31]

$$\max_{\xi^1,\ldots,\xi^T} \frac{1}{T}\sum_{t=1}^{T} \xi^t,$$

$$\text{s.t. } M \cdot (\xi^t + \xi^k - 2) \leq \kappa_{s'}^k - \kappa_s^t \quad \forall t,k \in T, s,s' \in S, q_s^t > q_{s'}^k \tag{HM-1}$$

$$\ln(\widehat{\pi}_s^t) + \kappa_s^t = \ln(p_s^t) + \gamma^t \quad \forall t \in T, s \in S \text{ and } q_s^t > 0 \tag{HM-2}$$

$$\ln(\widehat{\pi}_s^t) + \kappa_s^t \leq \ln(p_s^t) + \gamma^t \quad \forall t \in T, s \in S \text{ and } q_s^t = 0$$

When relating (HM-1) and (HM-2) to (RP-4) and (RP-3) it is useful to make the substitution $\kappa_s^t = \ln(\nu_s^t)$ and $\gamma^t = \ln(\lambda^t)$. The values of $\widehat{\pi}_s^t$ are given by Algorithm 2 (for a given μ).

The constant M should be large enough parameter such that if $\xi^t = 0$ or $\xi^k = 0$, then (HM-1) is automatically satisfied. Notice that in this case, one can always find a κ_s^t and γ^t that also satisfy (HM-2). On the other hand, if $\xi^t = \xi^k = 1$, then (HM-1) and (HM-2) are identical to logarized versions of (RP-3) and (RP-4). As such, the algorithm maximizes the size of the set $W = \{t \in T : \xi^t = 1\}$ that is CMEU rationalizable.

For the non-concave CMEU model in illustration 2 (where there are only 2 states), we define $q^{T+1} = (0,0)$, with $\xi^{T+1} = 1$.[32] The constraints are, $\forall t \leq T$ and $v, j \leq T+1$,

$$\begin{aligned}
&\text{If } p_1^t q_1^k + p_2^t q_2^j \leq p_1^t q_1^t + p_2^t q_2^t, \text{ then for } q = (q_1^k, q_2^j) \text{ we have} \\
&\quad M \cdot (\xi^t + \xi^k + \xi^j - 3) \leq \sum_{s \in S} \widehat{\Phi}(q^t)_s u(q_s^t) - \sum_{s \in S} \widehat{\Phi}(q)_s u(q_s) \\
&\text{If } p_2^t q_1^k + p_2^t q_2^j < p_1^t q_1^t + p_2^t q_2^t, \text{ then for } q = (q_1^k, q_2^j) \text{ we have that} \\
&\quad M \cdot (\xi^t + \xi^k + \xi^j - 3) < \sum_{s \in S} \widehat{\Phi}(q^t)_s u(q_s^t) - \sum_{s \in S} \widehat{\Phi}(q)_s u(q_s).
\end{aligned}$$

The constraints only bind if $\xi^t = \xi^k = \xi^j = 1$.

D Bingo Blower: 21 point grid analysis

Table 4 presents results for the Bingo Blower illustration where we increase the grid size of α and β. In particular, we now take 21 equally spaced values for α and 21 equally spaced values for β in their respective domains, and retain the values for which $\alpha + \beta \geq 0$. We

[31] A similar algorithm can be used to compute the HM-index for the SEU model.

[32] Remember that 0 has to be in the grid for the test to be valid.

see that increasing the number of grid points leads to a slight increase in the HM-index, which is expected. However, increasing the grid size does not qualitatively alter our results in the sense that, also on this finer grid, the SEU and CMEU model perform equally well.

Table 4: Average goodness of fit and Power

		Problem number					
		1	2	3	4	5	6
CMEU	HM-index	0.746	0.653	0.711	0.688	0.710	0.724
		(0.116)	(0.121)	(0.132)	(0.141)	(0.109)	(0.128)
	HM rand < HM	0.969	0.927	0.883	0.642	0.976	0.887
SEU	HM-index	0.745	0.652	0.709	0.688	0.709	0.722
		(0.118)	(0.122)	(0.132)	(0.141)	(0.110)	(0.129)
	HM rand < HM	0.967	0.928	0.881	0.642	0.976	0.886

Lines 2 and 4 give the average HM-index for the CMEU and the SEU models, standard deviations between brackets. Lines 3 and 5 give the average fraction of random datasets with an HM-index below the HM-index of the actual dataset.

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